

ORIGINAL RESEARCH

Targeting Thrombopoietin/MPL Signaling to Mitigate JAK2V617F-Driven Cardiac Microvascular Disease

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BACKGROUND: Individuals with *JAK2V617F*-mutant myeloproliferative neoplasms or clonal hematopoiesis of indeterminate potential have a markedly increased risk of cardiovascular disease, yet the mechanisms by which mutant blood cells drive vascular and cardiac dysfunction remain incompletely understood. Although the thrombopoietin receptor MPL is central to hematopoiesis and is expressed in vascular endothelial cells (ECs), its role in *JAK2V617F*-associated cardiovascular complications is unknown.

METHODS: We generated chimeric mice with *JAK2V617F*-mutant blood cells and wild-type endothelium by bone marrow transplantation and challenged them with a high-fat/high-cholesterol diet to model cardiometabolic stress.

RESULTS: Mice with *JAK2V617F*-mutant blood cells developed a distinct cardiovascular phenotype characterized by microvascular disease, increased left ventricular mass, and relatively preserved left ventricular ejection fraction. Histopathologic analysis revealed coronary arteriole stenosis, perivascular fibrosis, reduced microvascular density, and endocardial injury, without evidence of epicardial coronary stenosis or myocardial infarction. Single-cell RNA sequencing revealed activation of inflammatory, stress-response, and endothelial-to-mesenchymal transition gene signatures in ECs, most prominently within the endocardial ECs. Immunohistochemistry identified MPL expression predominantly in endocardial ECs. Thrombopoietin/MPL signaling was upregulated in endocardial ECs in mice with *JAK2V617F*-mutant hematopoiesis, and treatment with an anti-MPL neutralizing antibody markedly improved cardiovascular pathology, restored endocardial integrity, and increased coronary microvascular density.

CONCLUSIONS: *JAK2V617F*-mutant hematopoiesis induces cardiac microvascular dysfunction under cardiometabolic stress. Endocardial ECs play a critical role in this pathological process, and endocardial MPL signaling constitutes a potential targetable pathway in *JAK2V617F*-associated cardiovascular disease.

Key Words: endocardium ■ hematopoiesis ■ microcirculation ■ receptors ■ thrombopoietin

The acquired kinase mutation *JAK2V617F* drives clonal expansion of hematopoietic stem/progenitor cells and overproduction of mature, often dysfunctional blood cells. While it is best known as the driver mutation in myeloproliferative neoplasms (MPNs), accumulating evidence shows that *JAK2V617F* itself—rather than overt MPN—is a potent risk factor for cardiovascular disease (CVD). In patients with *JAK2V617F*-positive MPNs, CVD is the leading cause of morbidity

and mortality, with arterial or venous thrombosis affecting 40% to 50% of patients during the disease course, far exceeding age-matched controls.¹ *JAK2V617F* also occurs in clonal hematopoiesis of indeterminate potential (CHIP) without overt hematologic disease, present in ≈0.1% to 0.2% of middle-aged populations, with increased prevalence with aging.² Among all CHIP variants, *JAK2V617F* clones expand most rapidly³ and confer an ≈12-fold increased risk of coronary heart disease

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Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/ATVBAHA.126.325061>.

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Arterioscler Thromb Vasc Biol is available at www.ahajournals.org/journal/atvb

Nonstandard Abbreviations and Acronyms

CVD	cardiovascular disease
CHIP	clonal hematopoiesis of indeterminate potential
EC	endothelial cell
EndMT	endothelial-to-mesenchymal transition
HFD	high-fat/high-cholesterol diet
LV	left ventricular
LVEF	left ventricular ejection fraction
MPN	myeloproliferative neoplasm
scRNA-seq	single-cell RNA sequencing

What Are the Clinical Implications?

Individuals with *JAK2V617F*-mutant clonal hematopoiesis are known to have a markedly increased risk of cardiovascular disease, yet the underlying mechanisms remain incompletely understood. Our study establishes a clinically relevant murine model of cardiac microvascular disease driven by *JAK2V617F*-mutant hematopoiesis, implicating MPL-mediated endocardial endothelial dysfunction as a key pathogenic mechanism. These findings suggest that targeting MPL signaling to enhance the intrinsic repair and regenerative capacity of endocardial endothelial cells may represent a novel therapeutic strategy. This work has important clinical implications, as cardiac microvascular dysfunction is a major contributor to ischemia with no obstructive coronary artery disease and heart failure with preserved ejection fraction.

and ischemic stroke compared with those without any CHIP mutations—substantially higher than other CHIP mutations.^{4,5} Even small *JAK2V617F* clones (with allele burdens <2%) are associated with a >5-fold increase in cardiovascular events,⁶ underscoring the profound vascular impact of this mutation. Recent studies further demonstrate that *JAK2V617F*-mutant CHIP confers a 4-fold increased risk of heart failure independent of coronary artery disease, aligning with previous clinical observations that patients with *JAK2V617F*-positive MPNs have more than a 2-fold increased risk of developing heart failure.^{7,8} Together, these findings highlight a key unresolved question: how do *JAK2V617F*-mutant blood cells, whether in overt MPN or subclinical CHIP, promote vascular injury and cardiac dysfunction? Addressing this is essential for identifying modifiable pathways linking mutant hematopoiesis to CVD and for developing targeted risk-reduction strategies in this growing patient population.

Vascular endothelial cells (ECs) are central regulators of cardiovascular function. Thrombopoietin and its receptor MPL are best known for their roles in hematopoietic stem cell survival and megakaryocyte differentiation.⁹ Although MPL is mainly expressed on hematopoietic stem cells, megakaryocytes, and platelets,¹⁰ it is also present on vascular ECs—including human umbilical cord ECs,^{11,12} and mouse liver¹³ and pulmonary¹⁴ ECs—where thrombopoietin/MPL signaling modulates vascular function.^{12,13,15–17} While *JAK2V617F* is primarily expressed in hematopoietic cells in MPNs, it is also detectable in vascular ECs in some patients.^{18–20} We previously showed that mice expressing *JAK2V617F* in both hematopoietic cells and ECs develop spontaneous heart failure with thrombosis, vasculopathy, and reduced left ventricular ejection fraction (LVEF).²¹ Using EC-specific *JAK2V617F* mice and MPN patient-derived ECs, we demonstrated that *JAK2V617F* induces endothelial-to-mesenchymal transition (EndMT)²²—a process normally involved in cardiovascular development but that can be reactivated in postnatal life, either as an adaptive response to new

physiological demands or as a maladaptive response in disease.^{23–26} Notably, MPL expression is upregulated in *JAK2V617F*-mutant ECs compared with wild-type ECs,^{22,27,28} and genetic ablation of endothelial MPL suppresses *JAK2V617F*-induced EndMT and prevents CVD development.²²

Because *JAK2V617F*-mutant hematopoiesis is far more prevalent than endothelial mutation, we asked whether mutant blood cells alone can drive cardiovascular dysfunction and whether pharmacological blockade of thrombopoietin/MPL signaling with the neutralizing antibody AMM2^{29–34} can prevent these effects. In this study, we show that *JAK2V617F*-mutant blood cells induce endothelial dysfunction, evidenced by transcriptomic and histological abnormalities, even in the absence of detectable cardiac impairment. When challenged with a high-fat/high-cholesterol diet (HFD), however, these mice developed a distinct CVD phenotype characterized by cardiac microvascular disease with coronary arteriole stenosis, perivascular fibrosis, and endocardial disruption, accompanied by striking transcriptomic alterations in endocardial ECs revealed by single-cell RNA sequencing (scRNA-seq). Left ventricular (LV) mass was increased and systolic function measured by LVEF was relatively preserved. Endocardial ECs are a specialized population of ECs that form the innermost lining of the heart, covering the cardiac chambers and valves as part of the endocardium. We further discovered that MPL expression is selectively expressed in endocardial ECs, whereas arterial, venous, and capillary ECs were largely negative. Importantly, treatment with the MPL-neutralizing antibody AMM2 markedly ameliorated these diet-induced cardiovascular pathologies, restored endocardial integrity, increased coronary vascular density, and suppressed inflammatory and stress-response gene signatures in endocardial ECs. Together, these findings suggest that endocardial MPL signaling may represent a therapeutic

target to mitigate cardiovascular risk associated with JAK2V617F-mutant hematopoiesis.

METHODS

Data Availability Statement

RNA sequencing data have been deposited in the GEO data repository (GSE338171). Other data that support the findings of this study are available from the corresponding author (H. Zhan) on reasonable request.

Experimental Mice

JAK2V617F Flip-Flop (FF1) mice (which carry a Cre-inducible human JAK2V617F gene driven by the human JAK2 promoter; JAX stock 037558), Tie2-Cre mice (JAX stock 008863), and CD45.1 congenic mice (JAX stock 002014) were obtained from the Jackson Laboratory. FF1 mice were crossed with Tie2-Cre mice to express JAK2V617F in hematopoietic cells, generating Tie2-Cre⁺FF1⁺ mice on a CD45.2 background. All mice used were crossed onto a C57BL/6J background and housed in a pathogen-free mouse facility at Stony Brook University. Animal experiments were performed in accordance with the guidelines provided by the Institutional Animal Care and Use Committee.

Both male and female mice were included in the study, with data from both genders combined for analysis. Sex-specific differences and interactions were not evaluated in the study.

Isolation and Culture of Murine Cardiac ECs

Primary murine cardiac ECs (CD45⁺CD31⁺) were isolated from mouse hearts following a previously described protocol with minor modifications.^{21,22,27,28,35} Briefly, mice were euthanized, and the thoracic cavity was immediately opened via a midline sternotomy. Cardiac perfusion was performed with 20 mL cold PBS using a 25-gauge needle inserted into the LV apex, with concurrent drainage through a small incision in the right atrium. The heart was collected, rinsed 3× in PBS, and finely minced with scissors. The tissue fragments were then digested in DMEM medium containing 1 mg/mL Collagenase/Dispase (Roche, Switzerland) and 25 U/mL DNase (Sigma) at 37°C for 1 hour with gentle shaking. The digested tissue was then triturated using a 20 mL syringe attached to a 16-gauge needle for at least 12 passes to obtain a single-cell suspension, filtered through a 70 µm cell strainer, washed with 40 mL PBS, and centrifuged at 500g for 10 minutes. The resulting cell pellet was resuspended and sequentially processed for CD45⁺ cell depletion followed by CD31⁺ positive selection using magnetically labeled microbeads (Miltenyi Biotec, San Diego, CA) according to the manufacturer's protocol. Purified ECs (CD45⁺CD31⁺) were cultured on 2% gelatin-coated plates in complete EC medium.³⁵ No medium change was performed for the first 72 hours to allow EC attachment, followed by medium changes every 2 to 3 days.

In Vitro Cell Culture

For EC coculture with blood mononuclear cells, primary murine cardiac ECs (2×10⁴) were seeded onto 1% gelatin-coated 24-well plates and cultured in 500 µL of mouse complete EC

medium starting on Day −2. On Day 0, ECs were cocultured with 4×10⁴ JAK2 wild-type blood mononuclear cells (isolated from control mice) or JAK2V617F mutant blood mononuclear cells (isolated from Tie2-Cre⁺FF1⁺ mice) placed in Transwell inserts (1.0 µm pore size; Corning, NY) containing complete EC medium. After 2 days of coculture (Day 2), ECs were harvested, and their in vitro angiogenic capacities were assessed.

For the AMM2 treatment experiment, primary murine cardiac ECs (2×10⁴) were seeded onto 2% gelatin-coated 24-well plates and cultured in 500 µL of complete EC medium starting on Day −2. On Day 0, ECs were treated with either PBS or 100 ng/mL AMM2, an MPL-neutralizing antibody (Cat. 10401, IBL-America, Minneapolis, MN), for 48 hours. On Day 2, ECs were harvested, and their in vitro angiogenic capacities were assessed.

Endothelial Cell In Vitro Angiogenesis Assay

The EC tube formation assay was performed to evaluate in vitro angiogenic capacity.^{21,22,27,28,35} Matrigel matrix (Cat. 354234, Corning) was thawed overnight at 4°C and kept on ice until use. A prechilled 48-well culture plate was coated with 150 µL of Matrigel per well and incubated at 37°C for 30 minutes to allow gelation. 6×10⁴ primary mouse heart ECs in 300 µL of EC medium were seeded onto the solidified Matrigel. Tube formation was imaged at 2, 4, 6, 8, and 24 hours using a phase-contrast microscope (AMEX-1200, Thermo Fisher) at 4× and 10× magnifications. Quantitative analysis was performed using the ImageJ Angiogenesis Analyzer³⁶ (National Institutes of Health, Bethesda, MD) by counting the number of tubes and nodes in 20 to 30 nonoverlapping 400 × 400 pixel fields.

Stem Cell Transplantation, Diet, and AMM2 Treatment

Approximately 1×10⁶ marrow cells (either JAK2 wild-type cells from control mice or JAK2V617F-mutant cells from age-matched Tie2-Cre⁺FF1⁺ mice; CD45.2 background) were transplanted into lethally irradiated (two 540-cGy doses, 3 hours apart) 8-week-old wild-type recipient mice (CD45.1) via intravenous tail vein injection. Peripheral blood was collected every 4 weeks after transplantation to monitor complete blood counts and CD45.2⁺ donor chimerism.

For diet and AMM2 treatment: 8 weeks after transplantation, mice were assigned to one of the following groups for 5 weeks: (1) Standard diet: PicoLab Rodent Diet 20 5053; (2) HFD: TD88137 (Inotiv, Indianapolis, IN), containing 21% (wt/wt) milk fat and 49% (wt/wt) carbohydrate (sucrose and corn starch); (3) HFD plus AMM2 treatment: AMM2 (IBL-America 10401, Minneapolis, MN) administered intravenously at 0.1 mg/kg every 10 days for a total of 4 doses.

Mice were assigned to experimental groups using stratified randomization to ensure comparable male-to-female ratios across groups. Echocardiography was performed in a semi-blinded manner, as the sonographer measured a large number of animals and was not aware of individual treatment assignments at the time of data acquisition.

Complete Blood Counts

Peripheral blood was obtained from the facial vein via sub-mandibular bleeding and collected in EDTA-coated tubes. The

samples were then analyzed using a Vetscan Hm5 Hematology Analyzer (Abaxis) to determine the blood count.

Transthoracic Echocardiography

Transthoracic echocardiography was performed in mildly anesthetized, spontaneously breathing mice maintained under 1% to 3% isoflurane in oxygen at a flow rate of ≈ 1 L/min using a Vevo 3100 high-resolution imaging system (VisualSonics, Inc, Toronto, Canada), as previously described.^{21,22} Parasternal long-axis and sequential parasternal short-axis views were acquired to assess global and regional wall motion. LV end-systolic and end-diastolic dimensions were measured at the midventricular level, and LV ejection fraction and fractional shortening were measured using standard formulas to evaluate systolic function.³⁷

Histology

Hearts were fixed in cold 4% paraformaldehyde overnight at 4°C with gentle shaking, then washed several times with PBS at room temperature to remove residual paraformaldehyde. Paraffin-embedded sections (5 μ m thick) were stained with hematoxylin and eosin or Masson trichrome using standard protocols and reagents from Sigma (St. Louis, MO). Whole-slide images were acquired using a NanoZoomer S20 Digital Slide Scanner (Hamamatsu Photonics K.K., Japan), and representative images were captured with a Nikon Eclipse Ts2R Inverted Microscope.

Coronary microvasculopathy was assessed on hematoxylin and eosin-stained images using ImageJ (National Institutes of Health) as previously described.²² The ratio of luminal radius to wall thickness was quantified for each coronary arteriole (10–100 μ m in diameter), averaging 4 different measurements (top-bottom, left-right, and both diagonals). Arterioles were considered stenotic when the luminal radius-to-wall thickness ratio was <1 .^{38,39} To account for the variability in coronary arteriole morphology across treatment conditions, 6 to 8 mice were analyzed per group, with ≈ 70 to 100 microvessels evaluated in each group.

Perivascular and subendocardial collagen content was quantified on Masson trichrome-stained sections, where collagen fibers appeared blue, muscle fibers red, and nuclei black. For perivascular collagen quantification, intramyocardial vessels with circular profiles and intact morphology (10–100 μ m in diameter) were selected. The collagen-positive (blue) area surrounding each vessel was measured in ImageJ and expressed as a percentage of the total vessel area. For subendocardial collagen quantification, collagen content beneath the endocardial surface of both left and right ventricles was calculated as the ratio of collagen-positive subendocardial area to the length of the corresponding endocardial segment.

Single-Cell RNA Sequencing and Data Analysis

Single-cell suspensions from isolated mouse hearts were prepared by tissue mincing and enzyme digestion as described above in Isolation and Culture of Murine Cardiac Endothelial Cells, with the exception that the digestion step was shortened to 2 15-minute incubations (30 minutes in total). The digested tissue was gently triturated using a 20 mL syringe attached to a 16-gauge needle for at least 12 passes to obtain a single-cell

suspension, filtered through a 40 μ m cell strainer, washed with 40 mL PBS containing 2%FBS, and centrifuged at 500g for 10 minutes. Red blood cells were lysed 1 to 2 $\times$ using prewarmed Red Blood Cell Lysing Buffer (Cat. R7757, Sigma, Germany) with a 5-minute incubation at room temperature. After washing, cell pellets were resuspended in 100 μ L of Dead Cell Removal Microbeads per 10^7 total cells (Cat. 130-090-101, Miltenyi Biotec) and incubated for 15 minutes at room temperature. The mixture was diluted in 1 $\times$ binding buffer to a minimum volume of 500 μ L and applied to a Large Cell Column (Cat. 130-042-202, Miltenyi Biotec) mounted on a MACS Separator (Cat. 130-042-602, Miltenyi Biotec, Germany) to deplete dead cells bound to magnetic beads. Cells were washed twice with PBS containing 0.04% BSA and resuspended in PBS/0.04% BSA at a final concentration of 0.7 to 1.2×10^6 cells/mL.

Freshly isolated single-cell suspensions were immediately processed at the Single Cell Core Facility to generate scRNA-seq libraries using the Chromium Next GEM Single Cell 5' Reagent Kit v2 (10 $\times$ Genomics, Pleasanton, CA) according to the manufacturer's instructions. Briefly, cells, gel beads, and partitioning oil were loaded into a Chromium Next GEM Chip G for a target recovery of 10000 cells per sample. Gel Bead-in-Emulsions were generated on a Chromium Controller (10 $\times$ Genomics). Following reverse transcription using a Gel Bead-in-Emulsion reverse transcription incubation protocol, the Gel Bead-in-Emulsions were broken with Recovery Agent (10 $\times$ Genomics), and cDNA was purified using DynaBeads MyOne Silane beads (10 $\times$ Genomics), amplified by polymerase chain reaction (13 cycles), and purified using SPRIselect reagent (Beckman Coulter, Brea, CA). cDNA concentration and amplicon size were measured using a TapeStation 4200 with a D5000 high-sensitivity ScreenTape system (Agilent, Santa Clara, CA). After fragmentation, end repair, A-tailing, and size selection, cDNA was ligated with adapters and purified again using SPRIselect reagent, according to 10 $\times$ Genomics instructions. Libraries were amplified by polymerase chain reaction (10–11 cycles) using indexed oligonucleotide primers from the Chromium i7 Multiplex Kit (10 $\times$ Genomics). The final libraries underwent double-sided size selection with SPRIselect reagents and were quality-checked on a TapeStation 4200. cDNA libraries were sequenced on a NovaSeq X Plus (Illumina) using paired-end 150 reads (Read 1: 151 bp; i7 index: 10 bp; Read 2: 151 bp) at Novogene (Durham, NC).

Raw sequencing data FASTQ files were aligned to the mm10-2020-A mouse reference genome and processed with Cell Ranger 7.1.0 (10 $\times$ Genomics). Downstream analysis was performed using Seurat 5.2.0.⁴⁰ Ambient RNA contamination was corrected using SoupX,⁴¹ and doublets were identified and removed using DoubletFinder.⁴² Cells were filtered to include those with ≥ 300 detected genes (nGenes), ≥ 500 unique molecular identifiers, $<15\%$ mitochondrial gene content, and a log-normalized genes-to-unique molecular identifier ratio >0.8 . Data were normalized using NormalizeData with default parameters, and the top variable genes were identified using FindVariableFeatures (selection.method = vst and features = 3000). After scaling (ScaleData), dimensionality reduction was performed by principal component analysis, using the top 20 PCs as determined by an Elbow Plot. Data integration and clusters were visualized using 2D uniform manifold approximation and projection. Cluster annotation was based on canonical marker genes identified from published literature. Marker

expression was visualized in both uniform manifold approximation and projection and violin plots. Differentially expressed genes were identified using the FindMarkers function (Seurat) with adjusted *P* value ranking. Gene set enrichment analysis was performed using Gene Set Enrichment Analysis, v4.4.0.⁴³

Reverse Transcription Polymerase Chain Reaction Assays

CD45-CD31⁺ ECs were isolated from marrow, spleen, lung, and heart tissue single cells using magnetically labeled microbeads (Miltenyi Biotec, San Diego, CA) according to the manufacturer's protocol.^{21,22,27,28,35} Total RNA was isolated using the RNeasy Mini Kit (Qiagen) and was digested with RNase-free DNase (Qiagen). The RNA was then reverse transcribed to cDNA using the High-Capacity cDNA Reverse Transcription Kit (Thermo Fisher) following the manufacturer's protocol. The TaqMan Gene Expression Assay (Mm00440306_g1, Thermo Fisher) was used to detect mouse MPL gene expression in ECs.

Flow Cytometry

Cells were counted and stained with antibodies by incubating cell suspensions at 4°C for 30 minutes in the dark. Samples were analyzed using a CytoFLEX flow cytometer (Beckman Coulter, Indianapolis, IN). CD45 (Clone 30-F11; BioLegend, San Diego, CA), CD45.2 (Clone 104; BioLegend), CD45.1 (Clone A20; BD Biosciences, San Jose, CA), Gr-1 (Clone RB6-8C5; BioLegend), CD11b (Clone M1/70; BioLegend), CD45R/B220 (Clone RA3-6B2; BioLegend), CD3 (Clone 145-2C11; BioLegend), CD41 (Clone MWReg30; BioLegend), Lineage (Lin) cocktail (including CD3, B220, Gr1, CD11b, and Ter119; BioLegend), cKit (Clone 2B8; BioLegend), Sca1 (Clone D7; BioLegend), CD150 (Clone mShad150; eBioscience, San Diego, CA), CD48 (Clone HM48-1; BioLegend), CD31 (Clone 390; BD Biosciences), MPL (Clone AMM2; IBL-America), and Streptavidin PE (Cat. 405203; BioLegend) antibodies were used. Flow cytometry data were analyzed using Kaluza (Beckman Coulter).

Immunohistochemistry

Formalin-fixed, paraffin-embedded heart sections were baked at 60°C for 20 to 30 minutes in a hybridization oven to melt the paraffin, followed by sequential washes in xylene (3, 5, and 5 minutes) and graded ethanol (100%, 95%, 70%, and 50%; 3–5 minutes each) to remove paraffin and rehydrate the tissue. Slides were then rinsed under running tap water for 5 minutes.

Heat-induced antigen retrieval was performed by immersing slides in preheated Tris-EDTA buffer (pH 9) and heating in a steamer at 98°C for 25 minutes. Slides were cooled in the same buffer to room temperature (≥30 minutes), rinsed in TBST buffer for 5 minutes, and blocked with 1% BSA in TBST for 30 minutes at room temperature inside a humidity chamber.

Primary antibodies diluted in 1% BSA were applied and incubated overnight at 4°C in a humidity chamber: rabbit anti-mouse MPL (Ab232755, Abcam; 1:500), rabbit anti-mouse CD41 (Ab134131, Abcam; 1:800), rabbit anti-mouse CD31 (Ab182981; 1:600), and mouse anti-human/mouse alpha-smooth muscle actin (aSMA, Thermo Fisher; 1:2000).

After removing the primary antibody, slides were washed 3× in TBST (5 minutes each). Endogenous peroxidase activity was quenched with 0.6% hydrogen peroxide for 5 minutes at room temperature, followed by 3 additional TBST washes. Secondary antibody incubation was performed for 1 hour at room temperature using a goat anti-rabbit IgG HRP-linked antibody (Cat. 7074, Cell Signaling Technology; 1:100) diluted in 1% BSA.

Following 3 TBST washes, signal detection was performed using the AEC substrate kit (Abcam, ab64252) for 5 minutes for MPL, CD41, and CD31 staining and 1 minute for aSMA staining. After removing the AEC substrate, slides were rinsed twice with deionized water (3 minutes each), counterstained with Mayer hematoxylin (Cat. 786-1264, G-Biosciences) for 15 to 30 seconds, washed gently in running deionized water for 5 minutes, and treated in a bluing solution (10 g magnesium sulfate and 0.67 g sodium bicarbonate in 1 L of deionized water) for 30 seconds. After a final rinse in deionized water (5 minutes), slides were mounted with aqueous mounting medium (Cat. 27290, Cell Signaling Technology) and scanned using a NanoZoomer S20 Digital Slide Scanner (Hamamatsu Photonics K.K., Japan).

VE-Cadherin In Vivo Staining and Immunofluorescence Imaging of Heart Vasculature

Twenty micrograms of Alexa Fluor 647-conjugated monoclonal antibody that targets mouse VE-cadherin (clone BV13; BioLegend) was injected retro-orbitally into mice under anesthesia. Ten minutes after antibody injection, the mice were euthanized. Freshly harvested hearts were fixed in 4% paraformaldehyde in PBS for 6 hours at 4°C while rotating. The tissues were washed in PBS overnight to remove the fixative and cryoprotected in a 20% sucrose PBS solution at 4°C. The tissues were embedded in OCT (Tissue-Tek) and flash-frozen at −80°C.

To quantify coronary vascular density (VE-cadherin-positive endothelial area as a percentage of total tissue area), 5-μm-thick heart sections were imaged using a Nikon Eclipse Ts2R Fluorescent Microscope with a 20× objective. Image analysis was performed using ImageJ software (National Institutes of Health). First, images were converted to 32-bit grayscale. The detection threshold was adjusted to include all tissue while excluding empty areas, and the total tissue area was measured. Next, the images were processed using the FindEdges function, and the detection threshold was adjusted consistently across images to best isolate VE-cadherin signal from background autofluorescence, and the VE-cadherin-positive area was measured. Last, vascular density was calculated as (VE-cadherin-positive area)/(total tissue area) × 100.

Measurement of Peripheral Blood Plasma Cytokines

Peripheral blood was collected in an EDTA tubes, mixed gently by inverting the tubes 3 to 5×, and immediately centrifuged at 2000g for 10 minutes at 4°C. The resulting platelet-poor plasma was collected, aliquoted, and stored at −80°C until further analysis. Cytokine levels were measured using ELISA kits following the manufacturer's instructions: IL-1 beta (R&D

Systems, Cat. MLB00C), IL-6 (R&D Systems, Cat. M6000B), TNF-alpha (R&D Systems, Cat. MTA00B), TGF-beta 1 (R&D Systems, Cat. DB100C), thrombopoietin (R&D Systems, Cat. MTP00), and platelet factor 4 (R&D Systems, Cat. MCX400). All samples were analyzed in duplicate.

Quantification and Statistical Analysis

Statistical analyses were performed using GraphPad Prism 10 (San Diego, CA) and R software. All measurements were taken from distinct samples. Except for the scRNA-seq experiments, all assays were performed independently at least twice. Normality testing was not performed due to the relatively small sample sizes used in this study. ANOVA was used to assess differences in means among multiple groups, while an unpaired 2-tailed Student *t* test was applied to compare differences between 2 groups. Post hoc multiple-comparison tests were performed for all experiments analyzed by ANOVA. For in vivo experiments, randomization and blinding were not used. Sample sizes for animal studies were chosen to ensure a balance between the minimum numbers necessary

for efficient analysis and accounting for biological variability. Figures were generated using GraphPad Prism 10, R software, and BioRender.com.

RESULTS

JAK2V617F Mutant Hematopoiesis Induces Early-Stage CVD in Mice on a Regular Chow Diet

To study the impact of JAK2V617F-mutant blood cells on endothelial function, primary murine heart ECs (CD45-CD31⁺) were isolated from wild-type mice and cocultured with JAK2V617F-mutant blood cells for 48 hours. Tube formation in Matrigel, a measure of in vitro angiogenesis, was mildly but significantly enhanced in wild-type ECs after coculture with JAK2V617F-mutant blood cells compared with coculture with wild-type blood cells (Figure 1A).

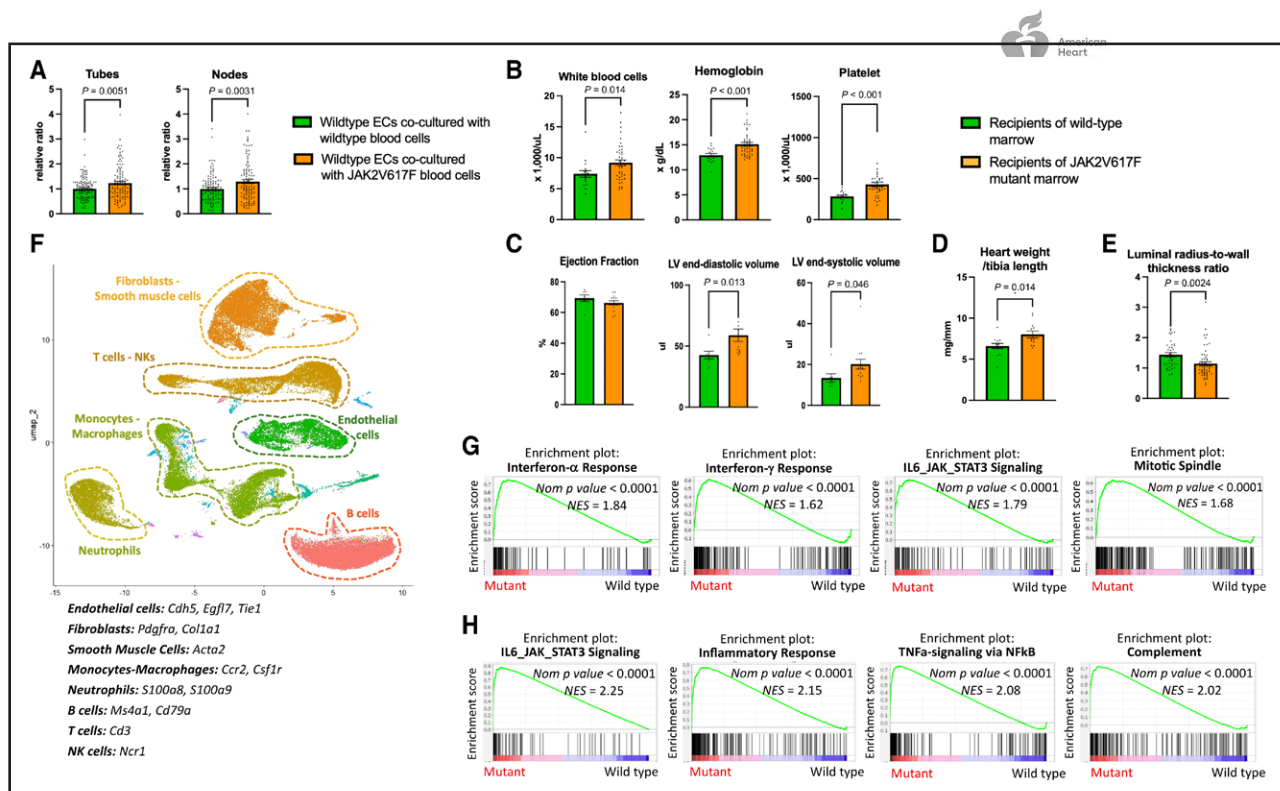


Figure 1. JAK2V617F-mutant hematopoiesis induces early-stage cardiovascular disease in mice on a regular chow diet.

A, Quantitative analysis showing a modest increase in tube formation in wild-type murine heart endothelial cells (ECs) cocultured with JAK2V617F-mutant blood cells compared with those cocultured with wild-type blood cells. Data shown are representative of 3 independent experiments that gave similar results. **B**, Peripheral blood counts demonstrating elevated white blood cell count, hemoglobin, and platelets in recipient mice transplanted with JAK2V617F-mutant marrow compared with wild-type marrow 8 weeks post-transplantation. **C**, Transthoracic echocardiography showing preserved left ventricular (LV) ejection fraction but increased LV end-diastolic volume and end-systolic volumes in recipients of JAK2V617F marrow compared with wild-type marrow 6 months after transplantation. $n = 7$ to 13 mice per group from 2 to 3 independent experiments. **D**, Heart weight normalized to tibia length in recipients of JAK2V617F marrow ($n = 12$) and wild-type marrow ($n = 13$) at 4 to 6 months post-transplantation. **E**, Luminal radius-to-wall thickness ratio measured in nonoverlapping coronary arterioles from recipients of JAK2V617F marrow and wild-type marrow (7–9 mice per group; 48–93 microvessels analyzed per group). **F**, Uniform manifold approximation and projection (UMAP) plot of unfraktionated cardiac cells, with key marker genes used to define major cell types. **G** and **H**, Gene set enrichment analysis of differentially expressed genes in cardiac ECs (**G**) and fibroblasts-smooth muscle cells (**H**) from mice with JAK2V617F-mutant vs wild-type blood cells. Data are presented as mean $\pm$ SEM, and *P* value was calculated using an unpaired 2-tailed Student *t* test.

To assess these effects *in vivo*, we transplanted JAK2V617F-mutant bone marrow (from Tie2-Cre⁺FF1⁺ mice, in which the mutation is expressed in all hematopoietic cells) or control marrow into lethally irradiated wild-type recipients, generating chimeric mice with mutant hematopoiesis and wild-type ECs. As expected, recipients of mutant marrow developed an MPN phenotype with elevated blood counts (Figure 1B). Although LVEF remained normal over 6 months by serial transthoracic echocardiography (66% versus 69%; $P=0.26$), LV end-diastolic (59 μ L versus 43 μ L; $P=0.013$) and end-systolic (20 μ L versus 13 μ L; $P=0.046$) volumes were increased in mutant mice compared with wild-type controls (Figure 1C). Although hemoglobin levels varied among the JAK2V617F-mutant mice (mean, 15.1 g/dL), LV ejection fraction did not differ between mice with below-average hemoglobin levels (mean, 13.6 g/dL) and those with above-average hemoglobin levels (mean, 17.7 g/dL; Figure S1). Consistently, mutant mice exhibited an increased heart weight-to-tibia length ratio at 4 to 6 months post-transplantation (Figure 1D). Coronary vascular analysis revealed mild arteriole narrowing in mutant mice (luminal radius-to-wall thickness ratio, 1.1 versus 1.4; $P=0.0024$; Figure 1E). In contrast to our prior murine models with combined endothelial and hematopoietic mutations, which showed prominent thrombosis in the lungs, right ventricles, and coronary arterioles,²¹ no overt thrombosis was observed in mice with mutant blood cells alone.

To further investigate how mutant blood cells alter cardiac ECs, we performed scRNA-seq on cardiac cells (excluding cardiomyocytes)⁴⁴ from recipients of mutant or wild-type blood cells $\approx$ 16 weeks post-transplantation (Figure 1F). Cardiac ECs (Cdh5⁺, Eglf1⁺, Tie1⁺) from mutant recipients showed significant enrichment of interferon- α/γ response, IL6-JAK-STAT3 signaling, and mitotic spindle pathways, consistent with endothelial inflammation and activation (Figure 1G). Taken together, these findings indicate that JAK2V617F-mutant blood cells induce early microvascular remodeling and vascular inflammation.

Before the onset of overt LV dysfunction, the heart often mounts a diffuse stress-response. ECs, which are in direct contact with circulating mutant blood cells, often release cytokines and growth factors that drive fibroblast and smooth muscle cell remodeling.⁴⁵ Consistent with this, fibroblasts (Pdgfra⁺, Col3a1⁺, Col1a2⁺, Col1a1⁺) and smooth muscle cells (Acta2⁺, Tagln⁺)⁴⁴ exhibited similar gene set enrichment analysis enrichment (ie, upregulation of IL6-JAK-STAT3 signaling, inflammatory response, and TNF α (tumor necrosis factor- α) signaling via NF κ B; Figure 1H), indicating a coordinated endothelial-mesenchymal program in mice with JAK2V617F-mutant blood cells. These changes were absent in monocytes/macrophages, neutrophils, T/NK cells, or B cells,

arguing against artifactual enrichment due to blood cell contamination.

JAK2V617F Mutant Hematopoiesis Promotes a Distinct Cardiac Microvascular Disease Phenotype Under a HFD

To test whether JAK2V617F-mutant blood cells promote CVD under metabolic stress, we treated the chimeric mice with mutant blood cells and wild-type ECs with either a standard diet or an HFD (TD.88137; Harlan Teklad). After 5 weeks, mutant mice exhibited reduced LV ejection fraction (59% versus 68%; $P=0.0011$) and increased LV end-diastolic (52 μ L versus 43 μ L; $P=0.028$) and end-systolic (21 μ L versus 14 μ L; $P<0.001$) volumes compared with controls (Figure 2A).

Pathological evaluation revealed an increased heart weight-to-tibia length ratio in mutant mice (Figure 2B). Mutant mice fed an HFD exhibited marked coronary arteriole stenosis (luminal radius-to-wall thickness ratio 0.9 versus 1.3; $P<0.001$; Figure 2C) and significantly increased perivascular collagen deposition (2.9-fold; $P<0.001$; Figure 2D). No gross atherosclerosis, epicardial coronary stenosis, or myocardial infarction was detected.

To investigate how HFD affects endothelial function in the presence of JAK2V617F-mutant blood cells, we performed scRNA-seq on unfractionated heart cells from mutant and wild-type mice after diet treatment. Cardiac ECs from HFD-treated mutant mice showed enrichment of EndMT, TNF α /NF κ B signaling, coagulation, P53, and inflammatory response pathways, indicating a shift toward a dysfunctional, proinflammatory, and mesenchymal/fibrotic state (Figure 2E). This endothelial reprogramming could explain the observed phenotype of impaired cardiac function, coronary arteriole stenosis, and perivascular fibrosis.

To further define the cellular players in JAK2V617F mutant blood cell-induced endothelial dysfunction, we performed an integrated cluster analysis of the EC population and identified 4 distinct EC populations: arterial ECs (Sema3g, Edn1, Nos3, Fbln5), venous ECs (Nr2f2, Lyve1, Igfbp5), capillary ECs (Rgcc, Aqp1, Cd36, Gpihbp1), and endocardial ECs (Nfatc, Nrg1, Tbx20, Cdh11)^{46,47} (Figure 2F). The most pronounced transcriptomic changes occurred in endocardial ECs from HFD-treated mutant mice, which showed robust upregulation of pathways involved in inflammation (eg, TNF α signaling, IFN α/γ signaling, inflammatory response), metabolic reprogramming (eg, MTORC1 signaling, fatty acid metabolism, oxidative phosphorylation), stress responses (eg, P53 pathway, apoptosis, hypoxia), and EndMT (Figure 2G). In line with these findings, histological evaluation revealed endocardial disruption—characterized by irregular lining, prominent nuclei, and cellular enlargement or thickening—in 7 of 8 mutant

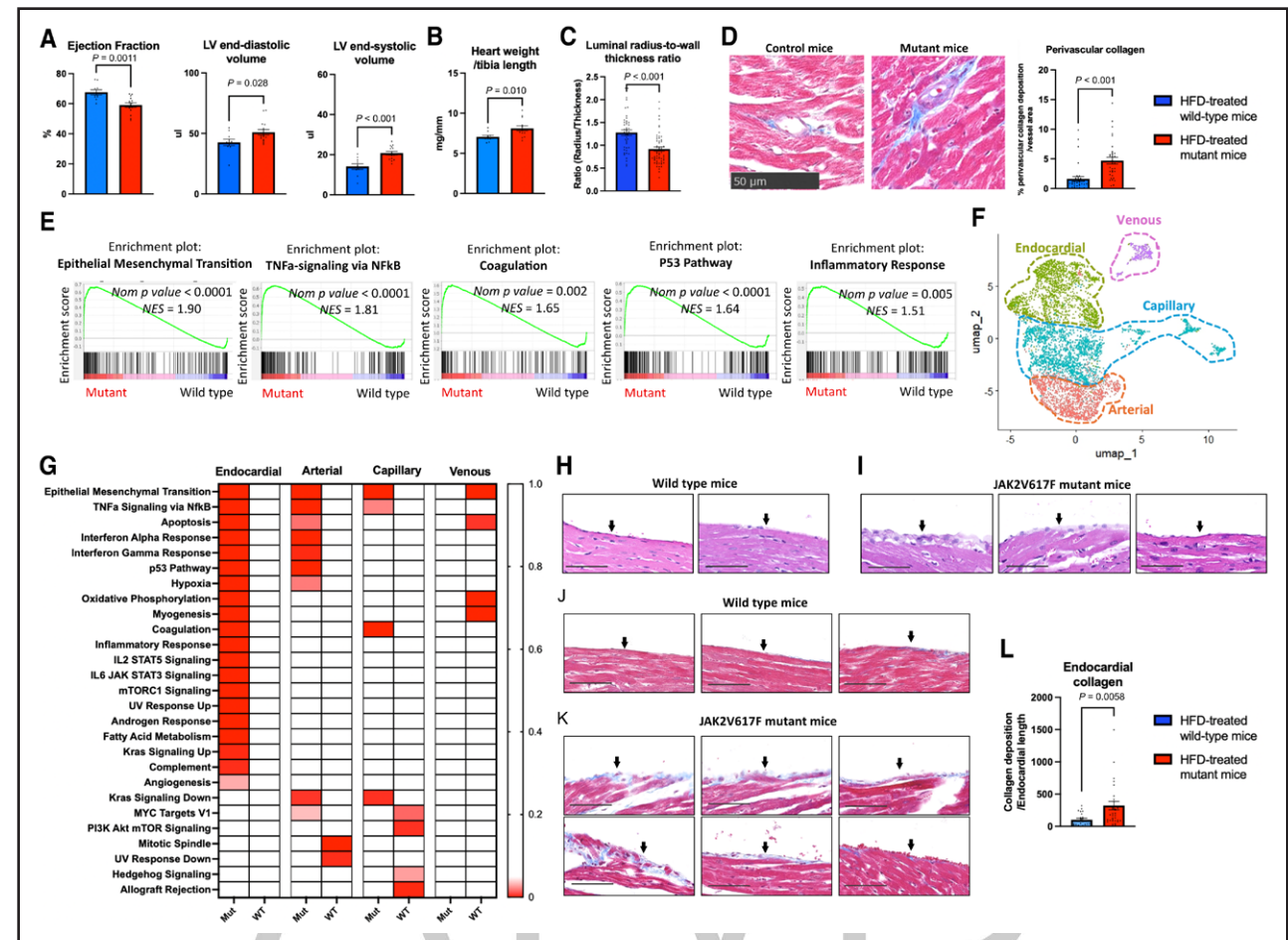


Figure 2. JAK2V617F-mutant hematopoiesis promotes a distinct cardiac microvascular disease phenotype under a high-fat diet challenge.

A, Transthoracic echocardiography showing reduced left ventricular (LV) ejection fraction and increased LV end-diastolic volume and end-systolic volumes in high-fat/high-cholesterol diet (HFD)-treated JAK2V617F-mutant mice (red) compared with wild-type controls (blue; $n=10-15$ mice per group from 3 independent experiments). **B**, Heart weight normalized to tibia length in JAK2V617F-mutant and wild-type control mice after 5 weeks of HFD treatment ($n=9-13$ mice per group; 3 independent experiments). **C**, Quantification of the luminal radius-to-wall thickness ratio, measured in nonoverlapping coronary arterioles from JAK2V617F-mutant mice and wild-type mice (6–8 mice per group; 72–100 microvessels analyzed per group). **D**, Representative Masson trichrome staining (left) and quantitation (right) of perivascular collagen deposition around coronary arterioles in JAK2V617F-mutant and wild-type mice (3 mice per group; 33 microvessels analyzed per group). Each dot represents an individual microvessel. **E**, Gene set enrichment analysis (GSEA) of differentially expressed genes in cardiac endothelial cells (ECs) from HFD-treated mice with JAK2V617F-mutant vs wild-type blood cells. **F**, Uniform manifold approximation and projection (UMAP) visualization of cardiac EC subpopulations. **G**, GSEA of endocardial, arterial, venous, and capillary ECs from HFD-treated JAK2V617F-mutant (Mut) vs wild-type mice. The color scale on the right indicates false discovery rate values. **H** through **K**, Representative hematoxylin and eosin (H and I) and Masson trichrome (J and K) staining of the endocardium (arrows) in HFD-treated wild-type and JAK2V617F-mutant mice. Scale bar: 50 μ m. **L**, Quantitation of subendocardial collagen deposition across nonoverlapping endocardium segments from JAK2V617F-mutant and wild-type mice (3 mice per group; 31–36 segments analyzed per group, average segment length 62 μ m). Data are presented as mean $\pm$ SEM and P value was calculated using an unpaired 2-tailed Student t test.

mice but in none of the 6 wild-type controls examined (Figure 2H and 2I). Masson trichrome staining showed significantly increased subendocardial collagen deposition in mutant mice (3.2-fold, $P=0.0024$; Figure 2J through 2L). Taken together, these findings indicate that JAK2V617F-mutant blood cells combined with HFD induced a distinctive CVD phenotype characterized by cardiac microvascular dysfunction, including coronary arteriole stenosis, perivascular fibrosis, and increased endothelial inflammation. LV mass is increased, whereas

LVEF remains relatively preserved. Notably, an unexpected yet consistent finding is disruption of the endocardial endothelium.

The endocardium is the innermost layer of the heart chambers and is composed of a thin monolayer of specialized ECs that line the cardiac interior. Endocardial cells are among the earliest endothelial populations to emerge during heart development and contribute to coronary vasculature formation during both embryogenesis⁴⁸ and postnatal life.⁴⁹ Given their critical role in

maintaining cardiac homeostasis and promoting repair and regeneration after cardiac injury,^{50,51} the impact of mutant blood cells on endocardial EC function warrants further investigation.

MPL Is Predominantly Expressed in a Subset of Endocardial ECs

The thrombopoietin receptor MPL is expressed on various vascular ECs,^{11–14} and thrombopoietin/MPL signaling has been implicated in cardiovascular regulation.^{12,13,15–17} While MPL expression has been reported in human umbilical cord ECs,^{11,12} mouse liver¹³ and pulmonary¹⁴ ECs, MPL expression within the heart has not been characterized. To address this, we examined MPL expression in murine cardiac ECs using multiple complementary approaches.

First, reverse transcription polymerase chain reaction was performed on ECs (CD45⁺CD31⁺) isolated from the heart, lung, marrow, and spleen of a wild-type C57BL/6J

mouse. cDNA generated by reverse transcription was amplified using primers targeting a 111-bp sequence spanning mouse MPL exons 2 to 3 (Mm00440306_g1, Thermo Fisher). MPL transcripts were consistently detected in ECs from all tissues (Figure 3A). Similarly, using a human MPL-specific primer, we detected MPL transcripts in human umbilical vein ECs, consistent with previous reports^{11,12} (data not shown).

Second, cell surface MPL protein expression was evaluated by flow cytometry analysis using the AMM2 antibody, a rat anti-mouse MPL neutralizing antibody (Cat. 10401, IBL-America). In line with our prior findings in MPL-GFP reporter mice (where GFP marks MPL transcriptional activity),^{22,33} MPL expression was highest in cardiac ECs, followed by lung ECs, and comparatively lower in marrow and spleen ECs (Figure 3B).

Third, we performed immunohistochemistry with a rabbit anti-mouse MPL antibody (ab232755, Abcam) to localize MPL expression within the heart's endothelial subtypes—endocardial, arterial, venous, and capillary.

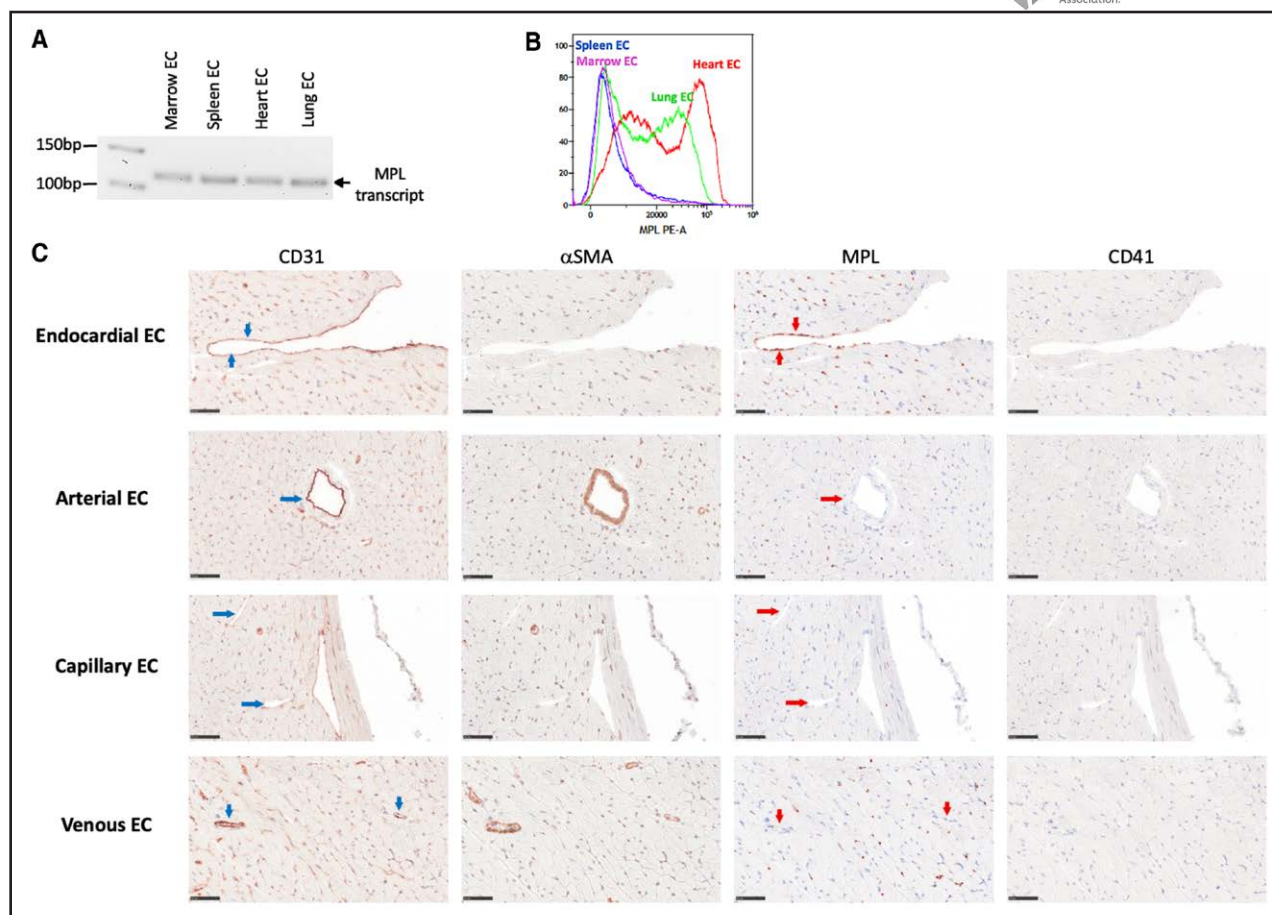


Figure 3. MPL is selectively expressed in a subset of endocardial endothelial cells (ECs).

A, MPL gene expression in isolated vascular ECs (CD45⁺CD31⁺) from heart, lung, bone marrow, and spleen of wild-type C57BL/6 mice, measured by reverse transcription-polymerase chain reaction. **B**, Cell surface MPL protein expression in ECs from heart, lung, bone marrow, and spleen, assessed by flow cytometry using the anti-MPL antibody AMM2. **C**, Representative immunohistochemical staining of mouse heart showing MPL localization within distinct endothelial subtypes. Blue arrows indicate vascular ECs with positive CD31 staining; red arrows denote MPL staining in the same cells, demonstrating positive expression in some endocardial ECs but not in arterial, capillary, or venous ECs. Scale bar: 50 μ m; magnification: 40 $\times$.

CD31 (or Pecam1) was used to identify vascular ECs, and CD41 to exclude platelets, which are MPL-positive. The endocardium was identified by its specific location lining the cardiac chambers and papillary muscles, whereas arteries were recognized by their thick media containing abundant concentric smooth muscle layers, veins by their thin walls with sparse smooth muscle, and capillaries by their single endothelial layer (no smooth muscle).³⁸ Strikingly, MPL expression was observed predominantly in a subset of endocardial ECs, whereas arterial, venous, and capillary ECs were largely negative (Figure 3C).

Previously, we showed that MPL expression is upregulated in JAK2V617F-mutant ECs compared with wild-type ECs,^{22,27,28} and that genetic ablation of endothelial MPL suppresses JAK2V617F-induced EndMT and prevents CVD development in mice with JAK2V617F-mutant ECs.²² Building on our new finding that MPL is selectively expressed in endocardial ECs, a subpopulation with critical roles in maintaining cardiac function and mediating repair and regeneration following cardiac injury,^{50,51} and that these cells exhibit marked transcriptomic and histological perturbations in mice with JAK2V617F-mutant hematopoiesis under metabolic stress (Figure 2), we next investigated whether pharmacological inhibition of thrombopoietin/MPL signaling using the MPL-neutralizing antibody AMM2 could prevent CVD development in vivo in mice bearing JAK2V617F-mutant blood cells.

AMM2 Prevents CVD Development in Mice With JAK2V617F Mutant Blood Cells by Restoring Endocardial Integrity and Microvascular Function

AMM2 was first described by Yoshihara et al²⁹ in 2007 as an anti-MPL neutralizing antibody, with a short course (6 days, 1 mg/kg) reducing cell-cycle regulators and increasing Lin⁺cKit⁺Sca1⁺ hematopoietic stem/progenitor cell cycling and engraftment in vivo. Olson et al³⁰ subsequently reported that AMM2 (1 mg/kg) administered at the time of irradiation did not reduce total marrow megakaryocytes, although it impaired their migration to the endosteal niche. Li et al³¹ demonstrated that AMM2 treatment (0.1 mg/kg per day for 4 days) enhanced the therapeutic effect of chemotherapy (Ara-C) in a leukemia murine model, likely by promoting leukemia stem cell cycling. In addition, AMM2 has been widely used as an MPL-specific antibody to measure receptor expression by flow cytometry and Western blot.^{32–34}

Since thrombopoietin/MPL signaling is a key regulator of HSC function,⁹ we first assessed whether AMM2 treatment would impair hematopoiesis. In a pilot study, wild-type mice received a single intravenous injection of AMM2 at 0.1 mg/kg. Peripheral blood platelet counts remained stable throughout a 14-day follow-up period (Figure 4A). We then treated wild-type mice with 0.1

mg/kg AMM2 intravenously every 10 days for a total of 4 doses to evaluate potential hematopoietic effects during extended treatment. No significant differences were observed in peripheral blood counts or in bone marrow megakaryocytes and Lin⁺cKit⁺Sca1⁺ hematopoietic stem and progenitor cells between AMM2- and PBS-treated mice (Figure 4B). Therefore, the 0.1 mg/kg AMM2 regimen was used for all subsequent in vivo experiments as a hematopoietically safe dose.

We generated chimeric mice harboring JAK2V617F-mutant blood cells and wild-type ECs through bone marrow transplantation. By 8 weeks post-transplantation, JAK2V617F-mutant mice had already developed a classic MPN phenotype, including leukocytosis, erythrocytosis, and thrombocytosis, compared with wild-type controls (as shown in Figure 1B).²¹ At this time, JAK2V617F-mutant mice were assigned to 1 of 2 groups for a total of 5 weeks of treatment: (1) HFD alone or (2) HFD plus AMM2, administered intravenously at 0.1 mg/kg every 10 days for a total of 4 doses. Chimeric mice with wild-type blood cells and ECs served as controls and received HFD alone (Figure 4C). Consistent with findings in wild-type mice, AMM2 treatment did not alter peripheral blood counts in JAK2V617F-mutant mice (Figure 4D) but provided clear cardiovascular protection: whereas mutant mice on HFD alone developed cardiac dysfunction (Figure 2A), AMM2-treated mice maintained normal LV ejection fraction and LV volumes comparable to those of wild-type controls (Figure 4E). The experiment was repeated 3×, each yielding consistent results. To ensure that the cardioprotective effect was not attributable to the antibody solvent (1% BSA and 0.05% sodium azide), we repeated the experiment using a customized AMM2 formulation without BSA and sodium azide (IBL-America) and observed identical results (data not shown). Histological analysis revealed that, compared with HFD treatment alone (Figure 2I and 2K), AMM2 markedly restored endocardial integrity in all HFD-treated JAK2V617F-mutant mice examined (n=6; Figure 4F). Quantitative assessment demonstrated persistent coronary arteriole stenosis; however, perivascular collagen fibrosis and subendocardial collagen deposition were significantly reduced in AMM2-treated mice (Figure 4G), suggesting that AMM2 improved both cardiac microvascular function and endocardial integrity.

To investigate the molecular mechanisms by which AMM2 prevents HFD-induced cardiovascular dysfunction in mice with JAK2V617F-mutant blood cells, we performed scRNA-seq on unfractionated heart cells from mutant mice on HFD and from mutant mice on HFD plus AMM2. AMM2 treatment markedly suppressed inflammatory (eg, TNFα/NFκB signaling, IFNα/γ signaling) and stress-response (eg, UV response, apoptosis, P53, hypoxia) pathways when all cardiac ECs were analyzed collectively (Figure 5A). Among the individual endothelial subtypes, endocardial ECs displayed the most

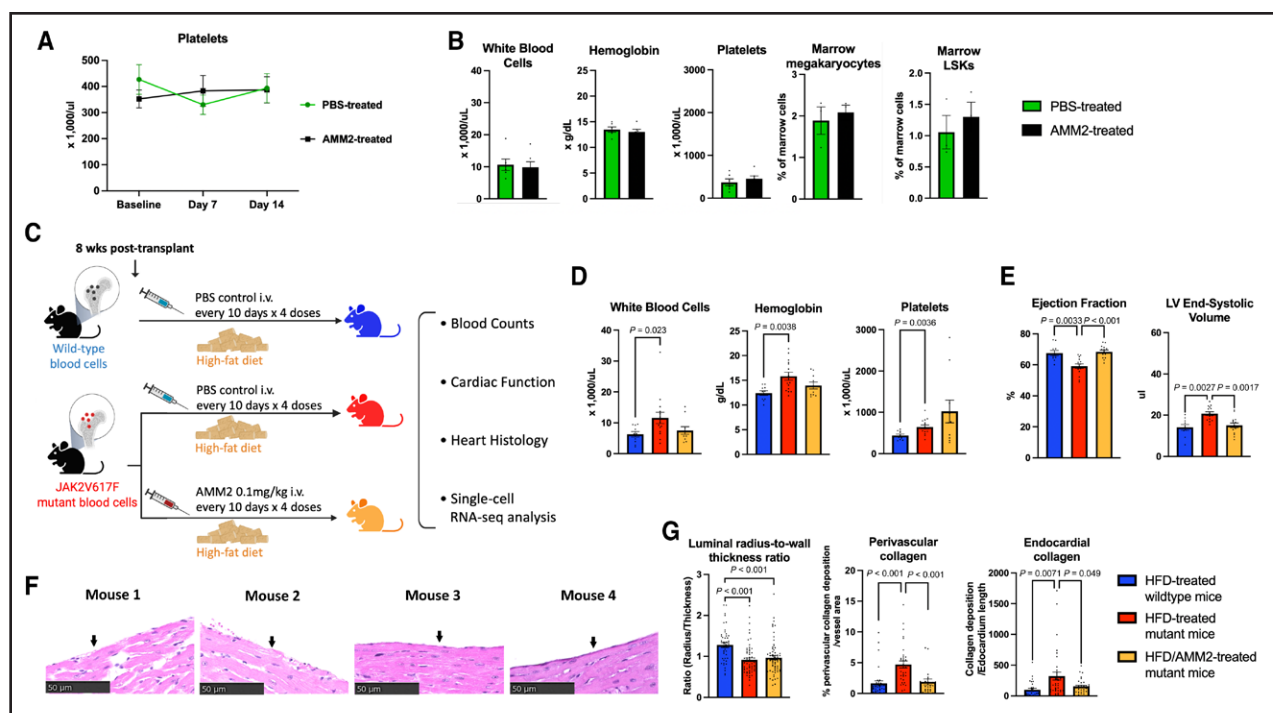


Figure 4. AMM2 prevents cardiac microvascular disease development in mice with JAK2V617F-mutant blood cells even under high-fat diet challenge.

A, Peripheral blood platelet counts in wild-type mice following a single dose of saline (PBS) or AMM2 ($n=4-7$ mice per group). **B**, Peripheral blood counts ($n=6$ per group) and bone marrow megakaryocyte and Lin⁻cKit⁺Sca1⁺ (LSK) hematopoietic stem and progenitor cells ($n=3$ mice per group) in wild-type mice after 4 doses of PBS or AMM2. **C**, Schematic illustration of the experimental design showing the high-fat/high-cholesterol diet (HFD) feeding and AMM2 treatment schedule. **D** and **E**, Peripheral blood counts (**D**) and left ventricular (LV) ejection fraction and volumes (**E**) in wild-type mice on HFD, JAK2V617F-mutant mice on HFD, and JAK2V617F-mutant mice on HFD+AMM2 ($n=10-15$ mice per group from 3 independent experiments). **F**, Representative hematoxylin and eosin staining of the endocardium (arrows) from 4 JAK2V617F-mutant mice treated with both HFD and AMM2. **G**, Quantification of (left) coronary arteriole luminal radius-to-wall thickness ratio (8 mice per group; a total of 84–100 microvessels analyzed per group), (middle) coronary arteriole perivascular collagen deposition (3 mice per group; a total of 33 microvessels analyzed per group), and (right) subendocardial collagen deposition (3 mice per group; a total of 31–36 segments analyzed per group, average segment length 62 μ m) in wild-type mice on HFD, JAK2V617F-mutant mice on HFD, and JAK2V617F-mutant mice on HFD+AMM2. * $P<0.05$, ** $P<0.01$, *** $P<0.001$, and **** $P<0.0001$. Data are presented as mean $\pm$ SEM, and P value was calculated using an unpaired 2-tailed Student t test (**A** and **B**) or 1-way ANOVA (**D**, **E**, and **G**).

pronounced transcriptomic changes (Figure 5B), which aligns with our finding that MPL expression is largely restricted to endocardial ECs (Figure 3C). These molecular changes, combined with the histological findings of enhanced endocardial integrity and reduced endocardial and perivascular fibrosis in AMM2-treated mice (Figure 4F and 4G), indicate that AMM2 treatment promotes a protective endothelial phenotype and minimizes cell death and maladaptive remodeling in HFD-treated JAK2V617F-mutant mice.

Consistent with chronic inflammation as a hallmark of JAK2V617F-mutant hematopoiesis,⁵² IL1 β (interleukin-1 beta), IL6 (interleukin-6), TGF β (transforming growth factor beta 1), and TNF α were significantly elevated in JAK2V617F-mutant mice compared with wild-type mice (Figure 5C); however, their levels were not affected by dietary intervention or AMM2 treatment (Figure S2). Therefore, while JAK2V617F-mutant hematopoiesis induced persistent systemic inflammation with elevated IL-1 β , IL-6, TGF β , and TNF α , AMM2 was

able to suppress inflammation in cardiac ECs, particularly within the endocardial EC population. Such attenuation of endothelial inflammation may shift the coronary microvasculature from a dysfunctional to a regenerative state by enhancing endothelial cell proliferation.⁵³ Supporting this model, in vivo VE-cadherin labeling revealed that, while HFD reduced coronary microvascular density in both wild-type and JAK2V617F-mutant mice, AMM2 treatment was able to increase coronary microvasculature in mutant mice (Figure 5D).

To investigate how AMM2 selectively targets endocardial ECs, we compiled a list of 130 thrombopoietin/MPL signaling-related genes by integrating multiple sources, including (1) the thrombopoietin signaling pathway from GeneGlobe (QIAGEN), (2) the BioCarta thrombopoietin pathway gene set, and (3) Bhat et al.⁵⁴ We then assessed pathway enrichment across various cardiac EC subpopulations using our scRNA-seq data set. Notably, endocardial ECs—but not arterial, capillary, or venous ECs—from mice carrying JAK2V617F-mutant blood cells exhibited

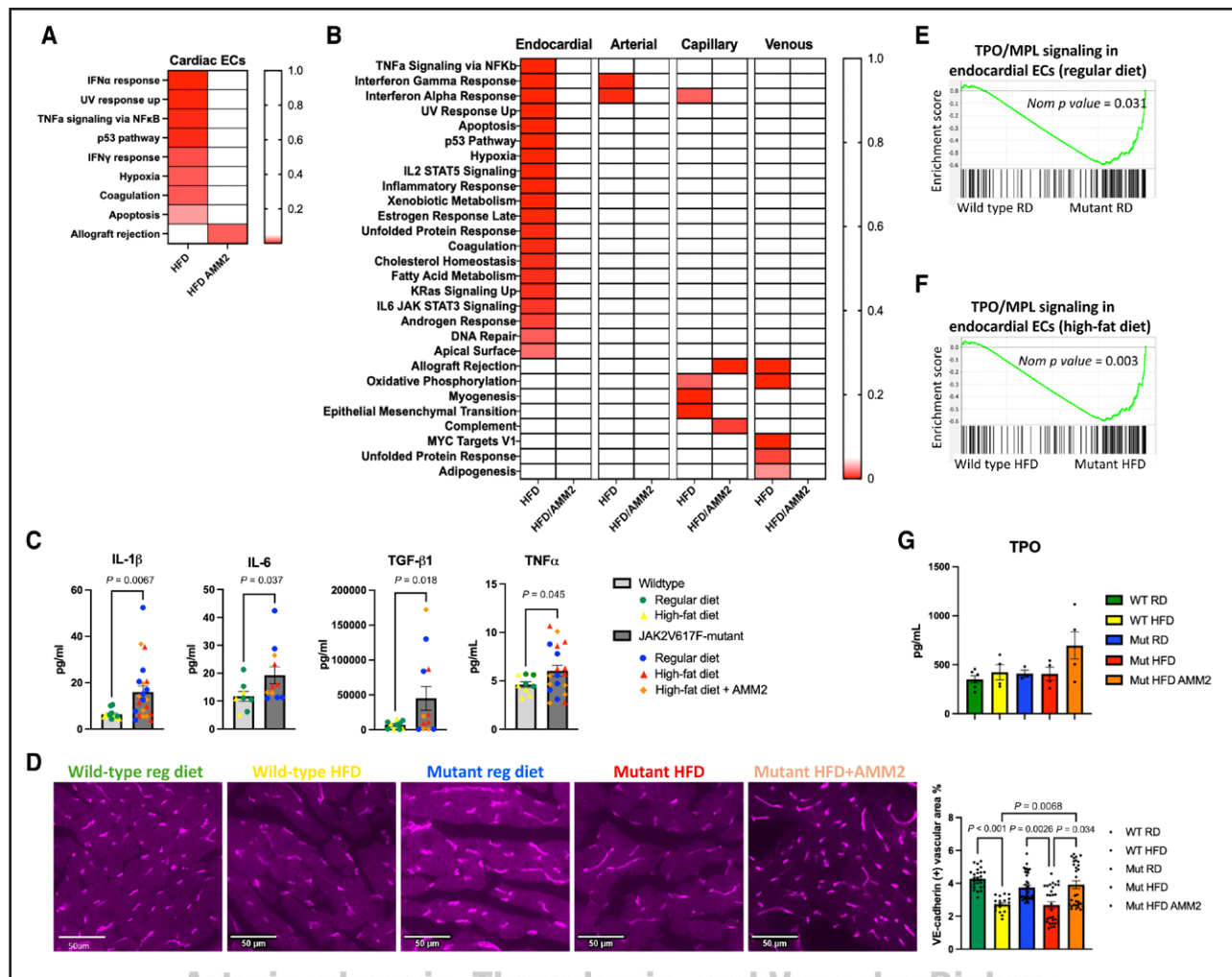


Figure 5. AMM2 treatment improves cardiac microvascular function in mice with JAK2V617F-mutant hematopoiesis despite persistent systemic inflammation.

A and **B**, Gene set enrichment analysis of differentially expressed genes in all cardiac endothelial cells (ECs; **A**) and EC subpopulations (**B**) from JAK2V617F-mutant mice treated with high-fat/high-cholesterol diet (HFD) vs HFD+AMM2. The color scale on the right indicates false discovery rate values. **C**, Plasma levels of IL1 β (interleukin-1 beta; $n=12-22$ mice per group), IL6 ($n=6-9$ mice per group), TGF β 1 (transforming growth factor beta 1; $n=10-12$ mice per group), and TNF α (tumor necrosis factor-alpha; $n=9-12$ mice per group) were significantly elevated in JAK2V617F-mutant mice compared with wild-type controls. **D**, Representative 20 $\times$ images and quantification of VE-cadherin $^{+}$ vasculature (magenta) in the hearts of wild-type (WT) and JAK2V617F-mutant (Mut) mice after 5 weeks of regular diet (RD), HFD, or HFD plus AMM2 treatment, demonstrating increased microvascular density in mutant mice treated with HFD+AMM2 compared with HFD alone ($n=2$ mice per group, with 19–30 20 $\times$ image fields analyzed per group). **E** and **F**, Thrombopoietin/MPL signaling genes were markedly upregulated in endocardial ECs from JAK2V617F-mutant mice compared with wild-type mice under both regular diet (**E**) and HFD (**F**) conditions. **G**, Plasma thrombopoietin levels across experimental groups ($n=3-5$ mice per group). * $P<0.05$, ** $P<0.01$, and **** $P<0.0001$. Data are presented as mean $\pm$ SEM, and P value was calculated using an unpaired 2-tailed Student t test (**C**) or 1-way ANOVA (**D** and **G**).

significant enrichment of thrombopoietin/MPL signaling genes compared with wild-type controls under both regular diet (Figure 5E) and high-fat diet (Figure 5F) conditions. Such changes were not observed in arterial, capillary, or venous cardiac ECs (Figure S3). Thrombopoietin/MPL signaling is primarily regulated by ligand-dependent activation through thrombopoietin binding. Although we did not detect significant differences in peripheral blood thrombopoietin levels among experimental groups (Figure 5G), these measurements may not reflect local cytokine dynamics within the cardiac microenvironment. Therefore, it is plausible that chronic inflammation associated with

JAK2V617F-mutant hematopoiesis, characterized by elevated cytokines such as IL-6,⁵⁵ enhances local thrombopoietin production or bioavailability within vascular ECs.¹³ Such paracrine or autocrine signaling could preferentially activate MPL in endocardial ECs, thereby driving the observed transcriptional program.

Together, these results identify endocardial ECs as an important target of JAK2V617F-mutant hematopoiesis and demonstrate that MPL is selectively expressed in this endothelial subset. The anti-MPL neutralizing antibody AMM2 alleviates stress (eg, HFD)-induced endocardial injury and cardiovascular dysfunction in

JAK2V617F-mutant mice, likely through suppression of endocardial inflammation and enhancement of coronary vascular repair and regeneration.

DISCUSSION

In this study, we show that mice harboring JAK2V617F-mutant blood cells develop a distinctive form of cardiac microvascular disease, both at baseline and after an HFD challenge, characterized by increased LV mass with preserved LVEF. Under HFD stress, a modest reduction in LVEF emerges, accompanied by clear evidence of endocardial endothelial dysfunction. Histologically, this phenotype is marked by coronary arteriole stenosis, perivascular fibrosis, endocardial disruption, and reduced microvascular density—hallmarks of cardiac microvascular dysfunction—in the absence of overt epicardial coronary artery disease.

Unlike prior studies using hyperlipidemic *Ldlr*^{-/-} mice⁵⁶ or invasive surgical models (eg, coronary ligation, transverse aortic constriction)⁵⁷ to link JAK2V617F-mutant blood cells to atherosclerosis and inflammation, our study employs a noninvasive high-fat/high-cholesterol Western diet model that represents a common and modifiable cardiovascular risk factor in developed countries.⁵⁸ A Western-style diet is a well-established preventable risk factor for CVD, and even a single high-fat meal can acutely impair endothelial function.^{59,60} Using this clinically relevant model, we show that JAK2V617F-mutant blood cells alone are sufficient to induce distinctive cardiac microvascular disease both at baseline and following an HFD challenge (Figures 1 and 2). In the adult heart, a tightly regulated interplay among ECs (including endocardial ECs), fibroblasts, and immune cells preserves microenvironmental homeostasis, enabling adaptation to stress and initiation of reparative responses.⁶¹ Our findings reveal that mutant hematopoiesis induces microvascular injury and triggers adaptive remodeling of the cardiac microenvironment, preserving LV systolic function via heightened endothelial inflammatory and stress signaling. Under metabolic stress (HFD), this compensatory response amplifies and increasingly depends on endocardial EC engagement. Prolonged HFD exposure ultimately disrupts this balance, leading to endocardial endothelial exhaustion, fibroblast activation, and maladaptive remodeling characterized by coronary arteriole stenosis, endocardial dysfunction, perivascular fibrosis, and increased LV volume/mass. These features define a pathophysiologic continuum leading to overt microvascular dysfunction in JAK2V617F-mutant mice under metabolic challenge. Unlike our prior model, which demonstrated overt heart failure with thrombosis, cardiomyopathy, and both macro- and microvasculopathy driven by mutant blood cells and mutant vascular ECs,²¹ this study offers a novel, noninvasive model of coronary microvascular disease.

Previous studies have linked JAK2V617F-mutant blood cells to vascular pathology: Poisson et al⁶² showed mutant erythrocytes induce endothelial oxidative stress and arterial vasoconstriction in the aorta and femoral arteries; Kimishima et al⁶³ reported worsened hypoxia-induced pulmonary hypertension with pulmonary arterial wall thickening; and Molinaro et al⁶⁴ demonstrated exacerbated endothelial damage and arterial thrombosis following carotid artery denudation. In contrast, our study focuses on cardiac ECs and provides several novel insights. We demonstrated for the first time that the thrombopoietin receptor MPL is selectively expressed in murine endocardial ECs, a population critical for coronary vasculature formation, cardiac repair, and regeneration.^{50,51} scRNA-seq identified these cells as the primary target of mutant hematopoiesis-driven CVD under cardiometabolic stress, showing activation of inflammatory, stress-response, and EndMT programs plus endocardial disruption. Importantly, the MPL-neutralizing antibody AMM2 ameliorated HFD-induced abnormalities by restoring endocardial integrity, increasing microvascular density, and downregulating inflammatory/stress-response signatures in endocardial ECs. Notably, low-dose AMM2 achieved these cardioprotective effects without altering peripheral blood counts in wild-type mice. Given the established role of MPL in MPN pathogenesis,^{65–71} our findings position endocardial ECs as a novel mediator of mutant hematopoiesis-driven cardiovascular injury and highlight cardiac endothelial MPL as a potential therapeutic target for preventing CVD in JAK2V617F-positive MPN or clonal hematopoiesis. Although we cannot exclude the possibility that increased blood viscosity in the MPN model may contribute to the inflammatory and stress-response gene signatures observed in cardiac ECs, LVEF did not differ between JAK2V617F-mutant mice with lower versus higher hemoglobin levels (Figure S1), suggesting that erythrocytosis alone is unlikely to account for the cardiac dysfunction in this model.

During embryonic development, endocardial ECs are essential for cardiac valve and septa formation through EndMT.⁷² They also contribute to coronary vasculature development embryonically^{48,73,74} and postnatally^{49,75–78} in zebrafish and mice, while supporting cardiac function, repair, and regeneration after injury.^{50,51} In the adult heart, EndMT is a process in which ECs lose canonical endothelial features and acquire mesenchymal/fibroblast-like properties. EndMT contributes to cardiac microvascular dysfunction through several mechanisms, including loss of endothelial integrity, decreased microvascular density (capillary rarefaction), perivascular fibrosis, and proinflammatory signaling, which were clearly demonstrated in our model. In our study, AMM2 treatment suppressed inflammatory and stress-response pathways in endocardial ECs, alleviated endocardial injury, increased coronary microvascular density, and improved cardiac function in HFD-treated JAK2V617F-mutant

mice, despite persistent systemic inflammation (elevated IL-1 β , IL-6, TGF β , and TNF α ; Figures 4 and 5). These findings suggest that endocardial EC-mediated coronary vascular repair/regeneration is a central mechanism of AMM2's cardioprotection. Although clonal hematopoiesis and associated systemic inflammation may be inevitable with aging, enhancing the intrinsic repair capacity of tissue-resident ECs offers a promising strategy to prevent aging-related cardiovascular diseases linked to clonal hematopoiesis. Further studies are required to understand the role of endocardial thrombopoietin/MPL signaling in JAK2V617F-induced cardiac microvascular disease and whether AMM2 acts directly on endocardial thrombopoietin/MPL signaling or indirectly via hematopoietic cells.

Our model may have clinical application by recapitulating some key pathological features of ischemia with nonobstructive coronary arteries and heart failure with preserved ejection fraction, in which coronary microvascular disease is increasingly recognized as a central pathogenic mechanism.^{79,80} Consistent with this framework, CHIP has been linked to increased prevalence of coronary microvascular dysfunction in patients with early, nonobstructive coronary artery disease.⁸¹ Large observational studies demonstrate that the association between CHIP and incident heart failure is independent of epicardial coronary artery disease, implicating non-ischemic, microvascular, or myocardial mechanisms.⁸² In a recent case-control study of incident heart failure, CHIP was associated with higher natriuretic peptide levels, with subgroup analyses indicating a stronger link to heart failure with preserved ejection fraction—particularly in individuals under 65 years—than to heart failure with reduced ejection fraction.⁸³ Moreover, in a large 2-cohort study, TET2-associated CHIP conferred more than a 2-fold increased risk of heart failure with preserved ejection fraction, independent of traditional cardiovascular risk factors and coronary artery disease.⁸⁴ Our model has provided an ideal platform for testing dietary/metabolic interventions and other therapeutic options, such as anti-inflammatory agents, to treat or prevent cardiac microvascular disease in clonal hematopoiesis.

In summary, our study has established a model in which JAK2V617F-mutant hematopoiesis drives cardiac microvascular injury and maladaptive remodeling, consistent with clinical observations in patients with clonal hematopoiesis. This process is amplified by metabolic stress and mediated, at least, in part, through selective vulnerability of endocardial ECs. We discovered MPL expression in endocardial ECs and showed that pharmacological MPL inhibition with AMM2 attenuates endocardial inflammation, preserves integrity, increases coronary vascular density, and improves cardiac function. These findings mechanistically link mutant hematopoiesis to human microvascular heart disease and highlight thrombopoietin/MPL signaling as a potential targetable

pathway to reduce cardiovascular risk in JAK2V617F-positive CHIP or MPNs.

ARTICLE INFORMATION

Received May 19, 2026; accepted August 14, 2026.

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Acknowledgments

The authors thank Ms. Isabel Zheng (Oldfield Middle School, Greenlawn, NY) for creating artwork for the graphic abstract.

Author Contributions

X. Yang performed all experiments, analyzed the data, and prepared the figures. K. Masarik contributed to the transgenic murine model studies and single-cell RNA sequencing data analysis. X. Sun conducted histology and plasma sample analysis and quantified vascular endothelial-cadherin-positive vascular areas. F. Zhang analyzed perivascular fibrosis and subendocardial collagen deposition. K. Zheng performed blinded quantification of VE-cadherin-positive vascular area. H. Zheng provided scientific and technical expertise in cardiovascular functional assessments and histological evaluations, interpreted the results, and drafted the manuscript. H. Zhan conceived and supervised the study, analyzed data, interpreted results, and drafted the manuscript. All authors reviewed and approved the manuscript.

Sources of Funding

This work was supported by the US Department of Veterans Affairs Merit Awards BX003947 and BX005584 (H. Zheng) and National Institutes of Health R01 CA266294 (H. Zhan).

Disclosures

None.

Supplemental Material

Figures S1–S3
Major Resources Table

REFERENCES

- Landolfi R, Di Gennaro L, Falanga A. Thrombosis in myeloproliferative disorders: pathogenetic facts and speculation. *Leukemia*. 2008;22:2020–2028. doi: 10.1038/leu.2008.253
- Bick AG, Weinstock JS, Nandakumar SK, Fulco CP, Bao EL, Zekavat SM, Szeto MD, Liao X, Leventhal MJ, Nasser J, et al; NHLBI Trans-Omics for Precision Medicine Consortium. Inherited causes of clonal haematopoiesis in 97691 whole genomes. *Nature*. 2020;586:763–768. doi: 10.1038/s41586-020-2819-2
- Pershad Y, Uddin MM, Xue L, Haessler J, Collins JM, Mack TM, Glick E, Glaser V, Zhao K, Jaiswal S, et al. Correlates and consequences of clonal hematopoiesis expansion rate: a 16-year longitudinal study of 6976 women. *Blood*. 2025;146:1078–1087. doi: 10.1182/blood.2025028417
- Jaiswal S, Fontanillas P, Flannick J, Manning A, Grauman PV, Mar BG, Lindsley RC, Mermel CH, Burt N, Chavez A, et al. Age-related clonal hematopoiesis associated with adverse outcomes. *N Engl J Med*. 2014;371:2488–2498. doi: 10.1056/NEJMoa1408617
- Jaiswal S, Natarajan P, Silver AJ, Gibson CJ, Bick AG, Shvartz E, McConkey M, Gupta N, Gabriel S, Ardissino D, et al. Clonal hematopoiesis and risk of atherosclerotic cardiovascular disease. *N Engl J Med*. 2017;377:111–121. doi: 10.1056/NEJMoa1701719
- Yokokawa T, Misaka T, Kimishima Y, Wada K, Minakawa K, Kaneshiro T, Yoshihisa A, Ikeda K, Takeishi Y. Clonal hematopoiesis and JAK2V617F mutations in patients with cardiovascular disease. *JACC CardioOncol*. 2021;3:134–136. doi: 10.1016/j.jacc.2021.01.001
- Flynn S, Schuermans A, Uddin MM, Nakao T, Viscosi V, Libby P, Natarajan P, Honigberg MC. Clonal hematopoiesis and incident heart failure. *JAMA Cardiol*. 2026;11:126–135. doi: 10.1001/jamacardio.2025.4603

8. Wu J, Zhang L, Vaze A, Lin S, Juhaeri J. Risk of Wernicke's encephalopathy and cardiac disorders in patients with myeloproliferative neoplasm. *Cancer Epidemiol*. 2015;39:242–249. doi: 10.1016/j.canep.2015.01.014
9. Kaushansky K. Thrombopoietin, the primary regulator of platelet production: from myths to logos, a thirty-year journey. *Biomolecules*. 2024;14:489. doi: 10.3390/biom14040489
10. Varghese LN, Defour JP, Pecquet C, Constantinescu SN. The thrombopoietin receptor: structural basis of traffic and activation by ligand, mutations, agonists, and mutated calreticulin. *Front Endocrinol (Lausanne)*. 2017;8:59. doi: 10.3389/fendo.2017.00059
11. Methia N, Louache F, Vainchenker W, Wendling F. Oligodeoxynucleotides antisense to the proto-oncogene c-mpl specifically inhibit in vitro megakaryocytopoiesis. *Blood*. 1993;82:1395–1401.
12. Brizzi MF, Battaglia E, Montrucchio G, Dentelli P, Del Sorbo L, Garbarino G, Pegoraro L, Camussi G. Thrombopoietin stimulates endothelial cell motility and neoangiogenesis by a platelet-activating factor-dependent mechanism. *Circ Res*. 1999;84:785–796. doi: 10.1161/01.res.84.7.785
13. Cardier JE, Dempsey J. Thrombopoietin and its receptor, c-mpl, are constitutively expressed by mouse liver endothelial cells: evidence of thrombopoietin as a growth factor for liver endothelial cells. *Blood*. 1998;91:923–929. doi: 10.1182/blood-2012-10-463414
14. English J, Dhanikonda S, Tanaka KE, Koba W, Eichenbaum G, Yang W-L, Guha C. Thrombopoietin mimetic reduces mouse lung inflammation and fibrosis after radiation by attenuating activated endothelial phenotypes. *JCI Insight*. 2024;9:e181330. doi: 10.1172/jci.insight.181330
15. Eguchi M, Masuda H, Kwon S, Shirakura K, Shizuno T, Ito R, Kobori M, Asahara T. Lesion-targeted thrombopoietin potentiates vasculogenesis by enhancing motility and enlivenment of transplanted endothelial progenitor cells via activation of Akt/mTOR/p70S6kinase signaling pathway. *J Mol Cell Cardiol*. 2008;45:661–669. doi: 10.1016/j.jmcc.2008.08.002
16. Lupia E, Spatola T, Cuccurullo A, Bosco O, Mariano F, Pucci A, Ramella R, Alloati G, Montrucchio G. Thrombopoietin modulates cardiac contractility in vitro and contributes to myocardial depressing activity of septic shock serum. *Basic Res Cardiol*. 2010;105:609–620. doi: 10.1007/s00395-010-0103-6
17. Li K, Sung RY, Huang WZ, Yang M, Pong NH, Lee SM, Chan WY, Zhao H, To MY, Fok TF, et al. Thrombopoietin protects against in vitro and in vivo cardiotoxicity induced by doxorubicin. *Circulation*. 2006;113:2211–2220. doi: 10.1161/CIRCULATIONAHA.105.560250
18. Sozer S, Fiel MI, Schiano T, Xu M, Mascarenhas J, Hoffman R. The presence of JAK2V617F mutation in the liver endothelial cells of patients with Budd-Chiari syndrome. *Blood*. 2009;113:5246–5249. doi: 10.1182/blood-2008-11-191544
19. Rosti V, Villani L, Riboni R, Poletto V, Bonetti E, Tozzi L, Bergamaschi G, Catarsi P, Dallera E, Novara F, et al. Associazione Italiana per la Ricerca sul Cancro Gruppo Italiano Malattie Mieloproliferative (AGIMM) investigators. Spleen endothelial cells from patients with myelofibrosis harbor the JAK2V617F mutation. *Blood*. 2013;121:360–368. doi: 10.1182/blood-2012-01-404889
20. Helman R, Pereira WO, Marti LC, Campregheer PV, Puga RD, Hamerschlag N, Chiattoni CS, Santos Fabio Pires de S. Granulocyte whole exome sequencing and endothelial JAK2V617F in patients with JAK2V617F positive Budd-Chiari Syndrome without myeloproliferative neoplasm. *Br J Haematol*. 2018;180:443–445. doi: 10.1111/bjh.14327
21. Castiglione M, Jiang YP, Mazzeo C, Lee S, Chen J-S, Kaushansky K, Yin W, Lin RZ, Zheng H, Zhan H. Endothelial JAK2V617F mutation leads to thrombosis, vasculopathy, and cardiomyopathy in a murine model of myeloproliferative neoplasm. *J Thromb Haemost*. 2020;18:3359–3370. doi: 10.1111/jth.15095
22. Zhang H, Kafeti N, Nasarik K, Lee S, Yang X, Zheng H, Zhan H. Decoding endothelial MPL and JAK2V617F mutation: insight into cardiovascular dysfunction in myeloproliferative neoplasms. *Arterioscler Thromb Vasc Biol*. 2024;44:1960–1974. doi: 10.1161/ATVBAHA.124.321008
23. Bischoff J. Endothelial-to-mesenchymal transition. *Circ Res*. 2019;124:1163–1165. doi: 10.1161/CIRCRESAHA.119.314813
24. Cho JG, Lee A, Chang W, Lee MS, Kim J. Endothelial to mesenchymal transition represents a key link in the interaction between inflammation and endothelial dysfunction. *Front Immunol*. 2018;9:294. doi: 10.3389/fimmu.2018.00294
25. Dejana E, Hirschi KK, Simons M. The molecular basis of endothelial cell plasticity. *Nat Commun*. 2017;8:14361. doi: 10.1038/ncomms14361
26. Li Y, Lui KO, Zhou B. Reassessing endothelial-to-mesenchymal transition in cardiovascular diseases. *Nat Rev Cardiol*. 2018;15:445–456. doi: 10.1038/s41569-018-0023-y
27. Zhan H, Lin CHS, Segal Y, Kaushansky K. The JAK2V617F-bearing vascular niche promotes clonal expansion in myeloproliferative neoplasms. *Leukemia*. 2018;32:462–469. doi: 10.1038/leu.2017.233
28. Lin CH, Kaushansky K, Zhan H. JAK2(V617F)-mutant vascular niche contributes to JAK2(V617F) clonal expansion in myeloproliferative neoplasms. *Blood Cells Mol Dis*. 2016;62:42–48. doi: 10.1016/j.bcmd.2016.09.004
29. Yoshihara H, Arai F, Hosokawa K, Hagiwara T, Takubo K, Nakamura Y, Gomei Y, Iwasaki H, Matsuoaka S, Miyamoto K, et al. Thrombopoietin/MPL signaling regulates hematopoietic stem cell quiescence and interaction with the osteoblastic niche. *Cell Stem Cell*. 2007;1:685–697. doi: 10.1016/j.stem.2007.10.020
30. Olson TS, Caselli A, Otsuru S, Hofmann TJ, Williams R, Paolucci P, Dominici M, Horwitz EM. Megakaryocytes promote murine osteoblastic HSC niche expansion and stem cell engraftment after radioablative conditioning. *Blood*. 2013;121:5238–5249. doi: 10.1182/blood-2012-10-463414
31. Li H, Zhao N, Li Y, Xing H, Chen S, Xu Y, Tang K, Tian Z, Wang M, Rao Q, et al. c-MPL is a candidate surface marker and confers self-renewal, quiescence, chemotherapy resistance, and leukemia initiation potential in leukemia stem cells. *Stem Cells*. 2018;36:1685–1696. doi: 10.1002/stem.2897
32. Ghinassi B, Zingariello M, Martelli F, Lorenzini R, Vannucchi AM, Rana RA, Nishikawa M, Migliaccio G, Mascarenhas J, Migliaccio AR. Increased differentiation of dermal mast cells in mice lacking the Mpl gene. *Stem Cells Dev*. 2009;18:1081–1092. doi: 10.1089/scd.2008.0323
33. Ng AP, Kauppi M, Metcalf D, Hyland CD, Josefsson EC, Lebois M, Zhang JG, Baldwin TM, Di Rago L, Hilton DJ, et al. Mpl expression on megakaryocytes and platelets is dispensable for thrombopoiesis but essential to prevent myeloproliferation. *Proc Natl Acad Sci U S A*. 2014;111:5884–5889. doi: 10.1073/pnas.1404354111
34. Nishikawa S, Arai S, Masamoto Y, Kagoya Y, Toya T, Watanabe-Okochi N, Kurokawa M. Thrombopoietin/MPL signaling confers growth and survival capacity to CD41-positive cells in a mouse model of Ev1 leukemia. *Blood*. 2014;124:3587–3596. doi: 10.1182/blood-2013-12-546275
35. Zhan H, Ma Y, Lin CH, Kaushansky K. JAK2V617F-mutant megakaryocytes contribute to hematopoietic stem/progenitor cell expansion in a model of murine myeloproliferation. *Leukemia*. 2016;30:2332–2341. doi: 10.1038/leu.2016.114
36. Carpentier G, Berndt S, Ferratge S, Rasband W, Cuendet M, Uzan G, Albanese P. Angiogenesis analyzer for ImageJ—a comparative morphometric analysis of “endothelial tube formation assay” and “fibrin bead assay”. *Sci Rep*. 2020;10:11568. doi: 10.1038/s41598-020-67289-8
37. Zaccigna S, Paldino A, Falcao-Pires I, Daskalopoulos EP, Dal Ferro M, Vodret S, Lesizza P, Cannata A, Miranda-Silva D, Lourenço AP, et al. Towards standardization of echocardiography for the evaluation of left ventricular function in adult rodents: a position paper of the ESC Working Group on Myocardial Function. *Cardiovasc Res*. 2021;117:43–59. doi: 10.1093/cvr/cvab110
38. Taqueti VR, Di Carli MF. Coronary microvascular disease pathogenic mechanisms and therapeutic options: JACC state-of-the-art review. *J Am Coll Cardiol*. 2018;72:2625–2641. doi: 10.1016/j.jacc.2018.09.042
39. Hiemann NE, Wellenhofer E, Knosalla C, Lehmkuhl HB, Stein J, Hetzer R, Meyer R. Prognostic impact of microvasculopathy on survival after heart transplantation: evidence from 9713 endomyocardial biopsies. *Circulation*. 2007;116:1274–1282. doi: 10.1161/CIRCULATIONAHA.106.647149
40. Hao Y, Stuart T, Kowalski MH, Choudhary S, Hoffman P, Hartman A, Srivastava A, Molla G, Madad S, Fernandez-Granda C, et al. Dictionary learning for integrative, multimodal and scalable single-cell analysis. *Nat Biotechnol*. 2024;42:293–304. doi: 10.1038/s41587-023-01767-y
41. Young MD, Behjati S. SoupX removes ambient RNA contamination from droplet-based single-cell RNA sequencing data. *GigaScience*. 2020;9:giaa151. doi: 10.1093/gigascience/giaa151
42. McGinnis CS, Murrow LM, Gartner ZJ. DoubletFinder: doublet detection in single-cell RNA sequencing data using artificial nearest neighbors. *Cell Syst*. 2019;8:329–337.e4. doi: 10.1016/j.cels.2019.03.003
43. Subramanian A, Tamayo P, Mootha VK, Mukherjee S, Ebert BL, Gillette MA, Paulovich A, Pomeroy SL, Golub TR, Lander ES, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc Natl Acad Sci U S A*. 2005;102:15545–15550. doi: 10.1073/pnas.0506580102
44. Skelly DA, Squiers GT, McLellan MA, Bolisetty MT, Robson P, Rosenthal NA, Pinto AR. Single-cell transcriptional profiling reveals cellular diversity and intercommunication in the mouse heart. *Cell Rep*. 2018;22:600–610. doi: 10.1016/j.celrep.2017.12.072
45. Cai H, Harrison DG. Endothelial dysfunction in cardiovascular diseases: the role of oxidant stress. *Circ Res*. 2000;87:840–844. doi: 10.1161/01.res.87.10.840
46. Kalucka J, de Rooij L, Goveia J, et al. Single-cell transcriptome atlas of murine endothelial cells. *Cell*. 2020;180:764–779.e20. doi: 10.1016/j.cell.2020.01.015

47. Peisker F, Halder M, Nagai J, Ziegler S, Kaesler N, Hoeft K, Li R, Bindels EMJ, Kuppe C, Moellmann J, et al. Mapping the cardiac vascular niche in heart failure. *Nat Commun*. 2022;13:3027. doi: 10.1038/s41467-022-30682-0
48. Wu B, Zhang Z, Lui W, Chen X, Wang Y, Chamberlain AA, Moreno-Rodriguez RA, Markwald RR, O'Rourke BP, Sharp DJ, et al. Endocardial cells form the coronary arteries by angiogenesis through myocardial-endocardial VEGF signaling. *Cell*. 2012;151:1083–1096. doi: 10.1016/j.cell.2012.10.023
49. Tian X, Hu T, Zhang H, He L, Huang X, Liu Q, Yu W, He L, Yang Z, Yan Y, et al. Vessel formation. De novo formation of a distinct coronary vascular population in neonatal heart. *Science*. 2014;345:90–94. doi: 10.1126/science.1251487
50. Brutsaert DL. Cardiac endothelial-myocardial signaling: its role in cardiac growth, contractile performance, and rhythmicity. *Physiol Rev*. 2003;83:59–115. doi: 10.1152/physrev.00017.2002
51. Zhang H, Lui KO, Zhou B. Endocardial cell plasticity in cardiac development, diseases and regeneration. *Circ Res*. 2018;122:774–789. doi: 10.1161/CIRCRESAHA.117.312136
52. Mendez Luque LF, Blackmon AL, Ramanathan G, Fleischman AG. Key role of inflammation in myeloproliferative neoplasms: instigator of disease initiation, progression, and symptoms. *Curr Hematol Malig Rep*. 2019;14:145–153. doi: 10.1007/s11899-019-00508-w
53. Evans CE, Iruela-Arispe ML, Zhao YY. Mechanisms of endothelial regeneration and vascular repair and their application to regenerative medicine. *Am J Pathol*. 2021;191:52–65. doi: 10.1016/j.ajpath.2020.10.001
54. Bhat FA, Advani J, Khan AA, Mohan S, Pal A, Gowda H, Chakrabarti P, KeshavaPrasad TS, Chatterjee A. A network map of thrombopoietin signaling. *J Cell Commun Signal*. 2018;12:737–743. doi: 10.1007/s12079-018-0480-4
55. Kaser A, Brandacher G, Steurer W, Kaser S, Offner FA, Zoller H, Theurl I, Widder W, Molnar C, Ludwiczek O, et al. Interleukin-6 stimulates thrombopoiesis through thrombopoietin: role in inflammatory thrombocytosis. *Blood*. 2001;98:2720–2725. doi: 10.1182/blood.v98.9.2720
56. Wang W, Liu W, Fidler T, Wang Y, Tang Y, Woods B, Welch C, Cai B, Silvestre-Roig C, Ai D, et al. Macrophage inflammation, erythrophagocytosis, and accelerated atherosclerosis in Jak2 (V617F) mice. *Circ Res*. 2018;123:e35–e47. doi: 10.1161/CIRCRESAHA.118.313283
57. Sano S, Wang Y, Yura Y, Sano M, Oshima K, Yang Y, Katanasaka Y, Min K-D, Matsuura S, Ravid K, et al. JAK2 (V617F)-mediated clonal hematopoiesis accelerates pathological remodeling in murine heart failure. *JACC Basic Transl Sci*. 2019;4:684–697. doi: 10.1016/j.jaccbts.2019.05.013
58. Sofi F, Abbate R, Gensini GF, Casini A. Accruing evidence on benefits of adherence to the Mediterranean diet on health: an updated systematic review and meta-analysis. *Am J Clin Nutr*. 2010;92:1189–1196. doi: 10.3945/ajcn.2010.29673
59. Collaborators GBDD. Health effects of dietary risks in 195 countries, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. *Lancet*. 2019;393:1958–1972. doi: 10.1016/S0140-6736(19)30041-8
60. Fewkes JJ, Kellow NJ, Cowan SF, Williamson G, Dordevic AL. A single, high-fat meal adversely affects postprandial endothelial function: a systematic review and meta-analysis. *Am J Clin Nutr*. 2022;116:699–729. doi: 10.1093/ajcn/nqac153
61. Tombar LS, Dimmeler S. Why is endothelial resilience key to maintain cardiac health? *Basic Res Cardiol*. 2022;117:35. doi: 10.1007/s00395-022-00941-8
62. Poisson J, Tanguy M, Davy H, Camara F, El Mdawar M-B, Kheloufi M, Dagher T, Devue C, Lassel J, Plessier A, et al. Erythrocyte-derived microvesicles induce arterial spasms in JAK2V617F myeloproliferative neoplasm. *J Clin Invest*. 2020;130:2630–2643. doi: 10.1172/JCI124566
63. Kimishima Y, Misaka T, Yokokawa T, Wada K, Ueda K, Sugimoto K, Minakawa K, Nakazato K, Ishida T, Oshima M, et al. Clonal hematopoiesis with JAK2V617F promotes pulmonary hypertension with ALK1 upregulation in lung neurophils. *Nat Commun*. 2021;12:6177. doi: 10.1038/s41467-021-26435-0
64. Molinaro R, Sellar RS, Vromman A, Sausen G, Folco E, Sukhova GK, McConke ME, Corbo C, Ebert BL, Libby P. The clonal hematopoiesis mutation Jak2(V617F) aggravates endothelial injury and thrombosis in arteries with erosion-like intimas. *Int J Cardiol*. 2024;409:132184. doi: 10.1016/j.ijcard.2024.132184
65. Pikman Y, Lee BH, Mercher T, McDowell E, Ebert BL, Gozo M, Cuker A, Wernig G, Moore S, Galinsky I, et al. MPLW515L is a novel somatic activating mutation in myeloid fibrosis with myeloid metaplasia. *PLoS Med*. 2006;3:e270. doi: 10.1371/journal.pmed.0030270
66. Pardanani AD, Levine RL, Lasho T, Pikman Y, Mesa RA, Wadleigh M, Steensma DP, Elliott MA, Wolanskyj AP, Hogan WJ, et al. MPL515 mutations in myeloproliferative and other myeloid disorders: a study of 1182 patients. *Blood*. 2006;108:3472–3476. doi: 10.1182/blood-2006-04-018879
67. Beer PA, Campbell PJ, Scott LM, Bench AJ, Erber WN, Bareford D, Wilkins BS, Reilly JT, Hasselbalch HC, Bowman R, et al. MPL mutations in myeloproliferative disorders: analysis of the PT-1 cohort. *Blood*. 2008;112:141–149. doi: 10.1182/blood-2008-01-131664
68. Sangkhae V, Etheridge SL, Kaushansky K, Hitchcock IS. The thrombopoietin receptor, MPL, is critical for development of a JAK2V617F-induced myeloproliferative neoplasm. *Blood*. 2014;124:3956–3963. doi: 10.1182/blood-2014-07-587238
69. Marty C, Pecquet C, Nivarthi H, El-Khoury M, Chachoua I, Tulliez M, Villeval JL, Raslova H, Kralovics R, Constantinescu SN, et al. Calreticulin mutants in mice induce an MPL-dependent thrombocytosis with frequent progression to myelofibrosis. *Blood*. 2016;127:1317–1324. doi: 10.1182/blood-2015-11-679571
70. Araki M, Yang Y, Masubuchi N, Hironaka Y, Takei H, Morishita S, Mizukami Y, Kan S, Shirane S, Edaheiro Y, et al. Activation of the thrombopoietin receptor by mutant calreticulin in CALR-mutant myeloproliferative neoplasms. *Blood*. 2016;127:1307–1316. doi: 10.1182/blood-2015-09-671172
71. Chachoua I, Pecquet C, El-Khoury M, Nivarthi H, Albu R-I, Marty C, Gryshkova V, Defour J-P, Vertenoel G, Ngo A, et al. Thrombopoietin receptor activation by myeloproliferative neoplasm associated calreticulin mutants. *Blood*. 2016;127:1325–1335. doi: 10.1182/blood-2015-11-681932
72. Eisenberg LM, Markwald RR. Molecular regulation of atrioventricular valvulogenesis morphogenesis. *Circ Res*. 1995;77:1–6. doi: 10.1161/01.res.77.1.1
73. Chen Q, Zhang H, Liu Y, Adams S, Eilken H, Stehling M, Corada M, Dejane E, Zhou B, Adams RH. Endothelial cells are progenitors of cardiac pericytes and vascular smooth muscle cells. *Nat Commun*. 2016;7:12422. doi: 10.1038/ncomms12422
74. Grego-Bessa J, Luna-Zurita L, del Monte G, Bojor V, Melgar P, Arandilla A, Garratt AN, Zang H, Mukoyama Y-S, Chen H, et al. Notch signaling is essential for ventricular chamber development. *Dev Cell*. 2007;12:415–429. doi: 10.1016/j.devcel.2006.12.011
75. Miquerol L, Thireau J, Bideaux P, Sturny R, Richard S, Kelly RG. Endothelial plasticity drives arterial remodeling within the endocardium after myocardial infarction. *Circ Res*. 2015;116:1765–1771. doi: 10.1161/CIRCRESAHA.116.306476
76. Zhao L, Borikova AL, Ben-Yair R, Guner-Ataman B, MacRae CA, Lee RT, Burns CG, Burns CE. Notch signaling regulates cardiomyocyte proliferation during zebrafish heart regeneration. *Proc Natl Acad Sci U S A*. 2014;111:1403–1408. doi: 10.1073/pnas.1311705111
77. Zhao L, Ben-Yair R, Burns CE, Burns CG. Endocardial Notch signaling promotes cardiomyocyte proliferation in the regenerating zebrafish heart through Wnt pathway antagonism. *Cell Rep*. 2019;26:546–554.e5. doi: 10.1016/j.celrep.2018.12.048
78. Kikuchi K, Holdway JE, Major RJ, Blum N, Dahn RD, Begemann G, Poss KD. Retinoic acid production by endocardium and epicardium is an injury response essential for zebrafish heart regeneration. *Dev Cell*. 2011;20:397–404. doi: 10.1016/j.devcel.2011.01.010
79. Tzouras A, Tapp D, Crousillat D, Honigberg MC, Rodriguez-Lozano PF, Aggarwal NR, Ebong IA, Briller J, De Oliveira GM, Harrington CM, et al; ACC Cardiovascular Disease in Women Committee. The role of coronary microvascular dysfunction in heart failure with preserved ejection fraction. *JACC Adv*. 2025;4:102345. doi: 10.1016/j.jaccadv.2025.102345
80. Rehan R, Yong A, Ng M, Weaver J, Puranik R. Coronary microvascular dysfunction: a review of recent progress and clinical implications. *Front Cardiovasc Med*. 2023;10:1111721. doi: 10.3389/fcvm.2023.1111721
81. Akhiyat N, Lasho T, Ganji M, Toya T, Shi C-X, Chen X, Braggio E, Ahmad A, Corban MT, Stewart K, et al. Clonal hematopoiesis of indeterminate potential is associated with coronary microvascular dysfunction in early nonobstructive coronary artery disease. *Arterioscler Thromb Vasc Biol*. 2023;43:774–783. doi: 10.1161/ATVBAHA.122.318928
82. Yu B, Roberts MB, Raffield LM, Zekavat SM, Nguyen NQH, Biggs ML, Brown MR, Griffin G, Desai P, Correa A, et al; National Heart, Lung, and Blood Institute TOPMed Consortium. Supplemental Association of Clonal Hematopoiesis with incident heart failure. *J Am Coll Cardiol*. 2021;78:42–52. doi: 10.1016/j.jacc.2021.04.085
83. Shi C, Aboumsallem JP, Suthahar N, de Graaf AO, Jansen JH, van Zeventer IA, Bracun V, de Wit S, Screever EM, van den Berg PF, et al. Clonal haematopoiesis of indeterminate potential: associations with heart failure incidence, clinical parameters and biomarkers. *Eur J Heart Fail*. 2023;25:4–13. doi: 10.1002/ehf.2715
84. Schuermans A, Honigberg MC, Raffield LM, Yu B, Roberts MB, Kooperberg C, Desai P, Carson AP, Shah AM, Ballantyne CM, et al. Clonal hematopoiesis and incident heart failure with preserved ejection fraction. *JAMA Netw Open*. 2024;7:e2353244–e2353244. doi: 10.1001/jamanetworkopen.2023.53244